Overlapping ownership in vertical relations: From fostering to tunneling

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The literature identifies several ways in which overlapping shareholders can reduce contractual and informational frictions between potential customers and suppliers. These effects, however, are compatible with overlapping owners also extracting private benefits from minority shareholders. I propose an instrumental variables approach to estimate the effect of more similar ownership structures separately from the effect of more concentrated ownership on either side of the relationship. My findings indicate that overlapping shareholders promote trading partnerships among companies within their portfolios, though the effect becomes four times larger when shareholder concentration differs substantially across companies. I argue that overlapping shareholders can create asymmetric incentives among managers, in which the diffuse-ownership company prioritizes the partner's gains while the concentrated company ignores potential losses for the other firm. Consistent with this mechanism, the same feature that makes a deal appealing to both parties can make it one-sided, fostering trading partnerships that enable tunneling along the supply chain.

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In the past, some have mistakenly assumed that our predominantly passive management style suggests a passive attitude with respect to corporate governance. Nothing could be further from the truth

William McNabb III
Vanguard's CEO, 2008-2018

The proliferation of large asset management institutions has accentuated the role of diversified minority shareholders in corporate strategy (Schmalz, 2018). In particular, with the rapid expansion of institutional financial intermediaries and a renewed preference for passive indexing strategies in the United States, a few institutional investors, such as Vanguard, BlackRock, and State Street, now rank among the top shareholders of most publicly listed companies in the country (Fichtner et al., 2017; Schmalz, 2018; Amel-Zadeh et al., 2022).[1]

The rise of large asset managers has led researchers to explore the implications of institutional investors holding stakes in two or more companies (henceforth, *overlapping shareholders*). As a result, a growing body of evidence now suggests that overlapping shareholders can soften competition (He and Huang, 2017; Azar et al., 2018), deter

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[1] Although the US stands out in the process, the trend is growing rapidly among large firms and certain geographic areas, such as Ireland, the Netherlands, and Sweden. Interestingly, recent evidence suggests that institutional ownership has been steadily high in regions like China, South America, and Mexico (Antón et al., 2025*b*).

entry (Xie and Gerakos, 2020; Newham et al., 2025), and produce supply-chain effects closely tied to the mechanisms of partial vertical integration, especially contractual and informational frictions (Deng et al., 2025; Freeman, 2025).

This paper builds on the finding that overlapping shareholders foster trading partnerships[2] among companies in their portfolios, and asks whether they may use these partnerships to extract private benefits at the expense of non-overlapping shareholders. Contrary to previous studies, my analysis focuses on mechanisms of partial vertical integration associated with asymmetric managerial incentives. That is, when executives across companies align to favor a single firm, something that existing identification strategies cannot reliably capture.

My findings suggest that the likelihood of a trading partnership increases as ownership structures among potential customers and suppliers become more similar, but disproportionately when shareholder concentration differs—a pattern consistent with managers on the side with more diffuse ownership using these partnerships to extract private benefits.

Several theoretical and empirical studies analyze the connection between the ownership structure of strategically interacting firms and market outcomes.[3] However, much of the literature still assumes that managers maximize the value of the company that appointed them. While reasonable at first glance, this premise overlooks that shareholders—who appointed managers in the first place—prioritize the total value of their portfolios rather than the individual value of each firm they own. Therefore, whether through direct investor influence or because managers internalize other benefits from pleasing pivotal shareholders, a company's ownership structure can alter its goals, market strategies, and business relationships.

In light of this, researchers have looked into the impact of *cross-ownership* and *common ownership* in product markets. While the former describes settings in which two companies hold shares in each other, the latter denotes third-party institutional shareholders across close competitors. Evidence from several studies documents that common ownership can hinder competition in several industries (Azar et al., 2018; Torshizi and Clapp, 2021; Koch et al., 2021; Newham et al., 2025), to the point that the European Commission has begun invoking common ownership as an element of competitive context in merger decisions (Burnside and Kidane, 2021).

Meanwhile, the role of overlapping owners in vertical relations remains somewhat underexplored. Nonetheless, if overlapping shareholders drive executives to weigh externalities imposed on competitors, why would managers ignore effects across upstream and downstream markets?

The distinction between horizontal and vertical overlapping ownership also invites researchers to reinterpret canonical insights about the separation of ownership and control (Jensen and Meckling, 1976). The literature on overlapping ownership departs from the traditional corporate finance standpoint, in which managers may pursue private benefits rather than maximizing firm profits (Jensen, 1986). Instead, it focuses on the idea that executives maximize the joint returns of shareholders' portfolios (Rotemberg, 1984).

Compared with horizontal settings, where companies and shareholders both benefit from softer competition, decisions along the supply chain can create winners and losers. Therefore, by tilting decisions toward the companies they hold most heavily, vertically

[2]Because customer and supplier identities matter, each pair of companies spans two different customer-supplier dyads.

[3]For example, several theoretical contributions argue that partial stakes in rival companies soften competition by shifting firm objectives away from own profits (Reynolds and Snapp, 1986; Bresnahan and Salop, 1986; Reitman, 1994; Gilo et al., 2006). For empirical evidence on the mechanism, see, for example, Parker and Roller (1997); Alley (1997); Nain and Wang (2018); Heim et al. (2022).

overlapping shareholders would shift the axis of conflict—from managers versus owners to overlapping versus non-overlapping shareholders—so that the gains from a trading partnership need not accrue to the companies that decide on it.

The paper centers on publicly listed companies in the United States between 1999 and 2013, drawing on several open data sources. I rely on prior research papers that compiled public filings, which companies must periodically submit to the Securities and Exchange Commission (SEC). Ownership ties—covering large institutional investors and blockholders—come from 13D, 13F, and 13G filings (Backus et al., 2021; Schwartz-Ziv and Volkova, 2025), which I use to assemble diverse overlapping ownership metrics following the theoretical approach in Rotemberg (1984). Likewise, I identify potential suppliers from the vertical upstream relatedness metric in Frésard et al. (2020) and active trading partnerships from 10-K filings (Barrot and Sauvagnat, 2016).

The analysis uses two-way fixed-effects regressions to examine the relationship between ownership structures and trading partnerships, combining observational weights with an instrumental variable approach to address two empirical challenges. First, companies will never establish infeasible trading partnerships, regardless of the strength of the mechanisms. Second, portfolio decisions that create overlap and managerial decisions on trading partnerships occur simultaneously.

I account for potentially infeasible trading partnerships in my sample by weighting each customer-supplier dyad by its vertical upstream relatedness. That is, the extent to which a supplier's output serves as an input for the customer's business. This approach renders unnecessary any assumption about the supply chain structure across industries, while still down-weighting dyads that are unlikely to trade.

My analysis compares results across multiple fixed-effects specifications and uses instruments constructed from fifteen years of panel data on shareholdings to address omitted variable bias. These instruments mirror the overlapping ownership measures used in the analysis but rely on predicted rather than observed shareholdings. Unlike prior identification strategies, which rely on exclusion restrictions based on unilateral decisions, my approach explicitly accounts for two-sided decisions: customers and suppliers jointly choose their trading partnerships.

Overall, increasing ownership-structure similarity by one standard deviation raises the likelihood of a trading partnership by roughly 10% to 19%—a magnitude comparable to Freeman (2025), who finds that overlapping owners reduce trading partnership terminations by roughly 26% to 32%.

Although prior studies contribute to the idea that overlapping shareholders help companies circumvent contractual and informational frictions (Riva, 2022; Freeman, 2025; Deng et al., 2025), my work points to an alternative explanation. Overlapping owners make trading partnerships much more likely when a company has a more diffuse ownership structure than its counterpart.

My empirical findings indicate that ownership-structure similarity has disparate effects on the likelihood of a trading partnership, depending on the relative shareholder concentration across customers and suppliers. For instance, when customers and suppliers have close to identical shareholder concentration, increasing ownership-structure similarity by one standard deviation has no statistically significant effect on the likelihood of a trading partnership. By contrast, when shareholder concentrations differ substantially, the same effect ranges from 28% to 84%.

In other words, in my preferred specification, the baseline effect roughly quadruples (from 19% to 84%) among customer-supplier dyads in which the relative shareholder concentration between the two companies deviates substantially from one. What drives such disproportionate effects when they stem from similar increases in ownership-structure similarity?

Throughout the paper, I argue that managers in firms with relatively diffuse ownership downplay outcomes within those firms, as gains and losses spread across many more minority shareholders.

Therefore, overlapping shareholders may bring about asymmetric managerial incentives, in which the diffusely owned company emphasizes its partner's gains while the concentrated company ignores potential losses for the other firm. Ultimately, when decisions mostly benefit one side of the trading partnership, managers pay closer attention to the company with more concentrated ownership, to the extent that the companies share investors.

My findings suggest that trading partnerships are more likely when overlapping shareholders induce asymmetric managerial incentives, creating opportunities to transfer equity value across companies and extract private benefits along the supply chain. Indeed, managerial decisions may result in direct losses or foregone profits, to the detriment of non-overlapping shareholders at the company that has more diffuse ownership.

This paper contributes to the common ownership literature, whose theoretical foundations date back to Rotemberg (1984), in which firms tend to act collusively to favor shareholders with more diversified portfolios. Even so, the topic gained traction and drew widespread attention when numerous researchers started reporting that overlapping shareholders among close competitors can soften competition and lead to welfare losses. For example, previous studies find that common ownership drives higher market concentration (He and Huang, 2017) and prices (Azar et al., 2018; Park and Seo, 2019; Torshizi and Clapp, 2021), negative effects on employment and wages (Azar et al., 2025), and reduced market entry (Xie and Gerakos, 2020; Newham et al., 2025).

Still, several papers document positive aspects related to overlapping ownership, such as spillovers in research and development activities (López and Vives, 2019; Antón et al., 2025*a*), increased product variety (Aslan, 2024), or faster innovation diffusion (Kostovetsky and Manconi, 2020). In particular, the few studies on vertical relations stand out for outlining how overlapping shareholders alleviate contractual and informational frictions. For example, firms secure larger loans at lower interest rates or are more likely to obtain syndicated loans when shareholders overlap (Ojeda, 2018). Similarly, they increase the likelihood of companies joining strategic alliances (Lindsey, 2008), discourage suppliers from engaging in upward earnings management (Gao et al., 2022), boost upstream partner-specific investments (Deng et al., 2025), and extend the lifetime of existing trading partnerships (Freeman, 2025).

I add to this literature in two different ways. First, my empirical work uncovers an adverse mechanism: asymmetric managerial incentives that lead executives to favor overlapping shareholders at the expense of non-overlapping ones. These findings add to a growing body of research warning of the risks of overlapping ownership. In particular, competition agencies have begun factoring in common ownership in their decisions, though they have done so for horizontal rather than vertical mergers.[4] While prior studies support the view taken by regulatory bodies by emphasizing efficiency gains akin to partial vertical integration, my work highlights a potential liability worth considering.

My second contribution concerns the empirical strategy, which circumvents the limitations posed by other sources of exogenous variation. For example, Berger (2023) challenges fire sales from mutual fund scandals, while Lewellen and Lowry (2021) critique the use of financial institution mergers and index additions for identification.

[4]In the EU, the European Commission invoked common shareholdings as an element of competitive context. For instance, see the European Commission Decision of 27 March 2017 in Case COMP/M.7932—*Dow/DuPont*, particularly paragraphs 2338 to 2352, and the European Commission Decision of 21 March 2018 in Case COMP/M.8084—*Bayer/Monsanto*, particularly paragraphs 976 to 981 and 3303 to 3306. In the US, see the FTC–DOJ Statement of Interest of 22 May 2025 in *Texas et al. v. BlackRock, State Street, Vanguard*, where plaintiffs plead common holdings as a potential source of standalone harm.

Although the literature has proposed other empirical strategies that work around earlier critiques, two reasons justify calling for new instruments in this paper, as none address the inherent challenges of this setting.

First, trading partnerships involve two-sided decisions by customer and supplier managers, whereas prior identification strategies targeted market outcomes based on unilateral company decisions, such as prices, entry, or investment. Thus, their exclusion restriction assumptions would require shifts in the ownership structure that did not affect other aspects of managerial decisions on either side of the supply chain relationship—a condition less likely to hold.

Second, prior identification strategies exploiting the addition of close competitors to indexes or ETFs (e.g., Antón et al., 2023; Newham et al., 2025; Azar et al., 2025) rely on systematic shifts in ownership-structure similarity across firms. While adequate for exploring symmetric effects on levels (such as market and firm outcomes), these empirical approaches fall short in capturing asymmetric managerial incentives. Identifying that variation requires an exogenous source of variation in shareholder concentration, which is what defines my design.

The remainder of the paper proceeds as follows. Section I describes the data, sample, and metrics of overlapping ownership. Section II discusses the empirical approach, particularly the use of weights in the analysis and the identification strategy. Section III shows that overlapping owners foster trading partnerships among companies within their portfolios, especially when tunneling opportunities exist. Finally, Section IV concludes.

I. Data: Ownership ties and the supply chain

The following section describes the data sources, the sample, and the metrics I use throughout the main analysis. For alternative definitions used in the robustness empirical exercises, see Section C in the Appendix.

I begin by introducing several open data sources on ownership and the supply chain and describing the sample construction. Next, I present two metrics that capture ownership patterns across companies: ownership-structure similarity and relative shareholder concentration.

A. Data

The analysis centers on publicly listed companies in the United States from 1999 to 2013, excluding those in the financial sector (SIC codes 6000-6999) and the transportation, communication, and utility sectors (SIC codes 4000-4999).

Publicly listed companies. I use *Compustat North America – Fundamentals*, which provides quarterly firm-level data, including outstanding shares, market value, closing price, net sales, total assets, and research and development (R&D) investment. In addition, I compute the company age from the first report of book assets since 1976 and include the Herfindahl-Hirschman Index from the text-based network industry classification (TNIC-HHI) by Hoberg and Phillips (2016). Furthermore, I winsorize R&D investment and total assets at the 1st and 99th percentiles and adjust nominal values for inflation to 2012 US dollars using the implicit price deflator from the Bureau of Economic Analysis.

Ownership ties. The SEC requires investment managers holding over USD 100 million in North American securities to file Form 13F quarterly, regardless of how many shares they own in a company. Blockholders, by contrast, must file Forms 13D and 13G once they acquire at least 5% of a public company's outstanding shares, and then annually thereafter.

For this paper, the US ownership network relies on two open data sources from prior research. I compile 13F filings submitted to the SEC between 1999 and 2017 from Backus et al. (2021) and combine them with 13D and 13G filings from 1998 to 2016 from Schwartz-Ziv and Volkova (2025).

Other SEC filings include Forms 3 to 5, which corporate insiders must file to inform ownership and transactions in company securities. They seldom coincide with institutional or blockholder positions, and are thus excluded from the analysis.

Trading partnerships. The supply chain network stems from the data published by Barrot and Sauvagnat (2016), covering trading partnerships among publicly listed companies in the United States and Canada from 1976 to 2013. Publicly traded companies file 10-K forms with the SEC every year and, under Statement of Financial Accounting Standards (SFAS) N°131, they must disclose the identity of any customer contributing at least 10% of total annual sales. This is a clear limitation in the data, as the wording of SFAS N°131 may lead to sample selection by making it less likely for small customers and large suppliers to exceed the mandatory 10% threshold.

In addition, *Compustat North America – Customer Segment* does not provide customer identifiers. Although prior studies have resorted to phonetic string-matching algorithms,[5] matching customers by disclosed names can introduce noise, as companies with similar names may refer to unrelated organizations despite shared historical roots.[6] Barrot and Sauvagnat (2016) extend the procedure by combining a phonetic string-matching algorithm with manual identification to recover the supply chain network.

Vertical upstream relatedness. I use the *vertical upstream relatedness* across company dyads to identify potential trading partners in the data. This directed measure appears in Frésard et al. (2020) and captures the degree to which a company produces outputs that another company uses as inputs. In practice, values close to one are extremely rare, as they would imply that every output of the supplier serves as an input to every good the customer produces. Thus, most company dyads have values close to zero, meaning that a trading partnership among them is practically unfeasible.

In summary, the authors map product descriptions from 10-K filings to a broad range of US commodities. Then, they combine this information with input-output intensities across commodities and construct directed measures for every pair of publicly listed companies in the US. Along with their paper, the authors have published annual updates from 1989 to 2023, which I employ to identify the subset of feasible customer-supplier dyads among all possible combinations of two companies.

B. Sample

The sample draws on the data sources described above, which overlap from 1999 to 2013. Panel A in Table 1 shows that roughly 7,500 companies are in business every year during this period, along with other relevant cross-year average characteristics.

However, the analysis covers a shorter window, spanning from 2000 to 2013. This restriction arises from my identification strategy, which requires a one-year lag in each company's ownership structure. For further details on the empirical approach, see Section II.

Moving forward, the paper focuses on the relationship between customers and suppliers, so that Panels B to E in Table 1 distinguish between companies that appear as

[5] See, for example, Fee et al. (2006), Wu and Birge (2014), or Cheung et al. (2020).

[6] For example, A.T. Massey Coal Company spun off Massey Energy Co. in 2000 and changed its name to Fluor Corp. Therefore, the string "Massey Coal Company" should match "Fluor Corp." rather than "Massey Energy Co.," which most phonetic string-matching algorithms would miss.

Table 1 – *Cross-year summary statistics for companies in Compustat*

	N	3-digit SIC industries	Age	Market value	Annual sales	Total assets	R&D intensity	TNIC-HHI	MB ratio
Panel A — Compustat									
All companies	7,374	222	11.44	2,854.45	2,329.67	1,860.12	26.00%	3,221	-0.41
Panel B — Feasible trading partners									
Customers	3,440	212	15.57	4,161.67	3,589.36	2,599.80	12.23%	3,224	2.65
Suppliers	3,472	213	15.40	3,499.50	2,770.85	2,213.27	12.45%	3,223	2.62
Panel C — Trading partnerships									
Customers	408	104	20.48	24,931.11	21,921.18	13,159.84	6.22%	2,563	3.25
Suppliers	889	132	15.26	3,618.89	2,332.86	2,137.56	13.24%	3,078	2.52
Panel D — Disclosing suppliers									
Customers	355	92	20.74	27,662.17	24,381.00	14,308.44	6.36%	2,520	3.29
Suppliers	664	112	15.92	3,649.98	2,377.55	2,140.80	13.21%	2,975	2.35
Panel E — Non-disclosing suppliers									
Customers	177	66	21.95	40,559.70	36,015.65	18,721.08	4.75%	2,588	3.65
Suppliers	225	77	13.49	3,491.98	2,221.38	2,107.96	13.61%	3,390	2.79

Notes: The table reports cross-year summary statistics for publicly listed US companies from 2000 to 2013. Panel A describes companies in industries other than the financial, transportation, communication, and utility sectors. Panels B and C focus on feasible customer-supplier dyads and trading partnerships observed in the data. Panel D restricts to suppliers that have disclosed at least one customer below the 10% threshold since 1976, along with their respective customers. Panel E, instead, centers on non-disclosing suppliers and their respective customers. The first and second columns indicate the average number of companies each year and the average number of 3-digit SIC industries they span, respectively. Market value, annual sales, and total assets are in millions of 2012 US dollars.

customers or suppliers in the data. Nonetheless, many publicly listed companies buy upstream and sell downstream, and therefore appear in both roles.

Feasible trading partners. Feasibility in trading partnerships hinges on what companies produce, not on whether they have an established relationship. For instance, Nokia announced in 2011 that it would ditch its flagship operating system, Symbian, and license Windows 7 and Bing from Microsoft. However, the agreement suggests that Microsoft could have been a potential supplier to Nokia even before 2011.

The challenge, though, is to capture all feasible trading partnerships, regardless of whether I observe them at any point in time. Otherwise, the analysis would assume technological constraints that make trading partnerships infeasible, when in reality it could reflect the endogenous choices of customers and suppliers.

In contrast, consider Activision Blizzard Inc., a video game holding company, and Alexion Pharmaceuticals Inc., a pharmaceutical and biotechnology company. Although both were publicly traded between 2005 and 2013, it is no surprise that the two companies did not establish a trading partnership, given the industries they operate in and the commodities they produce.

Infeasible customer-supplier dyads can bias my results because, regardless of their ownership structures, their products make them incompatible trading partners.[7] As a first step, I keep dyads with positive vertical upstream relatedness or an observed trading partnership.[8] Subsection II.B describes how using vertical upstream relatedness as observational weights helps address the bias, and Subsection C.1 in the Appendix

[7]Even determining the direction of such bias can be challenging, as the sign depends, among other things, on the ownership-structure similarity across feasible and infeasible firm dyads.

[8]The distinction stems from the fact that Frésard et al. (2020) lose data when mapping CUSIP (American Bankers Association) to GVKEY (Compustat) identifiers. Thus, supposedly infeasible customer-supplier dyads still appear in Compustat's Customer Segment.

shows that the conclusions remain unchanged when excluding trading partnerships that are supposedly infeasible but observed in the data.

The final sample contains 164,853,088 annual observations from 2000 to 2013, comprising 28,551,658 unique feasible trading partnerships between 6,018 customers and 6,113 suppliers. As mentioned above, the feasible set is drawn from 10-K product descriptions mapped to commodities, so companies with broader product descriptions are more likely to match with a counterparty. Consistent with this, customers and suppliers in Panel B have longer lifespans and lower R&D intensity, as well as higher sales, total assets, market values, and market-to-book ratios than the average firm in the Compustat dataset.

Trading partnerships. Following Barrot and Sauvagnat (2016), I define customer-supplier dyads as trading partnerships throughout all years between the first and last time the supplier reports the other firm as a significant customer in a 10-K filing. This procedure yields 23,723 trading partnerships between 2000 and 2013, comprising 5,823 unique dyads between 1,023 customers and 2,226 suppliers. Notice that trading partnerships are rare within the set of feasible customer-supplier dyads—roughly one in every seven thousand—and reported results must account for the small magnitudes.

Despite the wording of SFAS N°131, large suppliers do not appear underrepresented in the Customer Segment, given that cross-year average supplier characteristics do not differ significantly between Panels B and C. The problem arises among small customers because they make relatively smaller purchases and are less likely to surpass the 10% threshold for mandatory disclosure. Comparing cross-year average customer characteristics across Panels B and C shows that customers in the Customer Segment have a lifespan nearly 5 years longer, lower engagement in R&D activities, and significantly higher sales, market value, and total assets.[9]

Differences are more striking among customers purchasing from suppliers that appear stricter with the Statement. Although there is no obligation, many companies voluntarily disclose customers below the mandatory threshold. I classify suppliers by whether they have disclosed a customer below the 10% threshold at least once since 1976 (Panel D) or not (Panel E), with both panels displaying cross-year average characteristics for suppliers and their respective customers.

Panels B to E describe a problematic selection in the trading partnerships I observe in the Customer Segment—not only because the 10% threshold mechanically censors small customers, but also because many suppliers decide to disclose customers below the threshold anyway. I incorporate these concerns when designing my empirical strategy, which I detail in Section II.

C. Ownership-structure similarity and relative shareholder concentration

Overlapping ownership enters the analysis through two features associated with the interaction between ownership and corporate decisions: ownership-structure similarity and relative shareholder concentration across companies. In Rotemberg (1984), managers maximize a weighted average of the portfolios held by all the shareholders in the firm. The model highlights that investors do not need to actively engage with managerial decisions, but can still influence executives through perceived control and stock ownership. For a more detailed description of the framework and its derivation, see Section A in the Appendix.

According to the model, managers assign non-negative *profit weights*—often denoted by κ—to every company in the economy. Thus, the manager in company j weighs the

[9] For further discussion on the issue, see Barrot and Sauvagnat (2016) and Wu and Birge (2014). Both papers compare the Compustat - Customer Segment with alternative data sources, such as Capital IQ and Bloomberg SPLC.

effects of their decisions on the profits of company k in year t as follows

$$\kappa_{jkt} = \overbrace{\cos(\beta_{jt}, \beta_{kt})}^{\substack{\text{Ownership-structure} \\ \text{similarity (OSS)}}} \cdot \underbrace{\frac{\|\beta_{kt}\|}{\|\beta_{jt}\|}}_{\substack{\text{Relative shareholder} \\ \text{concentration (RSC)}}} \tag{1}$$

where β_{jt} depicts the ownership structure of company j in year t.[10] In addition, Equation (1) shows that κ can be written as the product of two interpretable terms that constitute the core of my analysis.

The first term in Equation (1) represents *ownership-structure similarity* (OSS), defined as the cosine similarity between the ownership structures of companies j and k, which ranges from 0 to 1.[11] A higher value in OSS indicates greater overlap among shareholders and more balanced ownership. For instance, consider two investors who initially hold shares in different companies. If they both sold their shares to a third investor with no prior holdings in either of them, the increase would reflect greater ownership overlap. By contrast, consider two overlapping shareholders, each with a stronger position in one company. If they exchanged shares proportionally to their portfolios, OSS would rise as shareholdings become more balanced.

The second term in Equation (1) captures *relative shareholder concentration* (RSC) across companies, defined as the ratio of the Euclidean norms of the corresponding ownership structures. The term gains its name because both the numerator and denominator match the square root of the Herfindahl-Hirschman Index of investor concentration for companies k and j, respectively. Thus, a higher shareholder concentration in the company that appoints the manager reduces RSC. Whereas a higher shareholder concentration in the firms whose profits are weighted increases RSC.

Profit weights prove compelling because they capture not only ownership overlap but also how shareholders can tilt executives' attention. In particular, each term governs a separate margin of how ownership interacts with managerial decisions in customer-supplier dyads.

OSS is a shape-only (scale-free) statistic that determines whether managers care about the other company's profits at all. Without shared investors, profit weights become zero, and managers turn indifferent to the externalities they may impose on potential trading partners.

RSC, by contrast, is a scale-only (shape-free) statistic that determines whether managerial attention is symmetrical. This is especially true because managers assign the same profit weights to each other only when shareholder concentrations are equal, regardless of OSS. However, a company whose owners are more concentrated can capture a disproportionate share of attention from its partner's manager, which is the exact mechanism through which concentrated companies can tunnel equity value from trading partners with diffuse ownership.

This reasoning finds a correlate in previous work. For instance, Backus et al. (2021) uses the framework in Rotemberg (1984) and concludes that profit weights above one imply tunneling incentives, as defined by Johnson et al. (2000), among executives. Along the same lines, Gilje et al. (2020) develop a metric for overlapping ownership that

[10] Note that β_{jt} is an $|\mathcal{S}|$-dimensional vector with elements in $[0, 1]$, representing the fraction of total shares held by investor $i \in \mathcal{S}$. Since minority and insider shareholdings are unobserved, the elements of β_{jt} do not necessarily add up to 1.

[11] Cosine similarity is the complement of the angular distance between two vectors in an inner product space, and it characterizes whether they point in roughly the same direction.

incorporates asymmetric managerial incentives through the internalization of voting rather than cash flow rights.

I combine all ownership ties from 13D, 13F, and 13G filings—excluding duplicated stakes where an investor files both 13F and 13D/G—and reconstruct the ownership network for US publicly listed companies between 1999 and 2013. Each entry indicates the fraction of outstanding shares owned by shareholders $s \in \mathcal{S}$ in company $j \in \mathcal{J}$. Following Equation (1), I employ these data to construct annual metrics of OSS and RSC, which I use throughout the main analysis. For a discussion about alternative metrics found in the literature and the corresponding robustness analysis, see Subsection C.2.

II. Empirical strategy

In this section, I define the empirical approach used throughout the paper. I begin by describing the empirical exercises that constitute the core of my analysis and the three fixed-effects specifications that help me account for the selection bias discussed in Subsection I.B. The following subsections detail the empirical challenges that arise from unobserved infeasibility and endogeneity, as well as my strategies to address them. I describe how observational weights, based on vertical upstream relatedness among dyads, can account for infeasible trading partnerships that remain in the sample. Second, I discuss my instrumental variables approach, which uses analogous versions of OSS and RSC based on predicted rather than observed shareholdings.

A. Estimating fostering and tunneling effects

Two empirical exercises sit at the core of the paper: overlapping shareholders foster trading partnerships, and the effect strengthens when shareholder concentration differs significantly between customers and suppliers. I estimate both effects using panel data regressions and a linear probability model, comparing estimates across different specifications to assess potential confounders when drawing conclusions.

The literature recognizes multiple reasons why certain firms are more likely to form trading partnerships (Bernard et al., 2019; Carvalho and Tahbaz-Salehi, 2019), and Table 1 shows that customers in observed trading partnerships tend to be larger companies. Combined with the censoring introduced by the wording of SFAS N°131, OSS and RSC estimates could pick up variation unrelated to the mechanisms I propose. To account for this, I develop three fixed-effects specifications, each addressing a different source of bias.

Time-varying company-FE ($\delta_{st} + \delta_{ct}$)**.** A natural response to potential confounders is to add time-varying covariates to the empirical model, such as firm age, total assets, annual sales, and industry concentration. This path, however, leaves the problem unresolved and introduces another potential identification threat. On the one hand, it does nothing about unobserved firm characteristics associated with the likelihood of trading partnerships. On the other hand, many covariates face the same simultaneity problems as ownership structure, and would act as *bad controls* when added directly to the empirical model (Angrist and Pischke, 2009).

The first specification includes time-varying company fixed effects, controlling for idiosyncratic trends across companies. It resembles two-way fixed effects (TWFE) estimation, while taking advantage of the fact that I can split the unit of analysis—the dyad—into customers and suppliers. Thus, the estimation still captures variation in the outcome and ownership across potential trading partners, as every company in the sample has multiple feasible customers and suppliers each year.

Dyad-FE and year-FE ($\delta_{sc} + \delta_t$)**.** This combination of fixed effects follows a conventional TWFE specification, with one flexibly controlling for aggregate time trends

and the other accounting for time-invariant idiosyncratic characteristics at the unit-of-analysis level. The upside is that dyad-FE holds constant several unobserved predictors of trading partnerships in the data, such as geographic proximity, relative company size, and other determinants associated with SFAS N°131 censoring. The downside is that the estimation captures only within-dyad variation in the outcome and ownership. Since the mechanisms I discuss insinuate that most variation lies between dyads, the coefficients from this specification may underestimate the effects.

Dyad-FE and time-varying company-FE ($\delta_{sc} + \delta_{st} + \delta_{ct}$)**.** The third specification combines the previous two, capturing within-dyad variation across customers and suppliers of every company. Thus, it is the most stringent of the three, accounting for most potential confounders but leaving the least variation for estimation.

Fostering. The first empirical exercise studies whether overlapping shareholders foster trading partnerships among companies in their portfolios. For that reason, I estimate the following empirical model

$$\text{LTP}_{sct} \;=\; \delta_{FE} \;+\; \tau \cdot \text{OSS}_{sct} \;+\; \varepsilon_{sct} \tag{2}$$

where LTP_{sct} denotes the *likelihood of a trading partnership*, defined by whether supplier s and customer c had a trading partnership during year t, and δ_{FE} represents one of the three fixed-effects specifications.

Given that trading partnerships are rare within the set of feasible customer-supplier dyads, I normalize LTP and OSS to facilitate interpretation. Therefore, τ reports the percentage change in the average probability of a trading partnership in the sample (roughly 0.02%) associated with a one-standard-deviation increase in OSS. I compute both averages using observational weights, as discussed in the next subsection.

Furthermore, I follow Angrist and Pischke (2009) and cluster standard errors at the dyad-level, where both LTP and OSS are measured. Importantly, the sample contains tens of millions of dyads, so the cluster-robust estimator does not suffer from small-cluster problems.

Tunneling. To explore the role of asymmetric managerial incentives, I extend the empirical model described in Equation (2) to account for differences in shareholder concentration across customers and suppliers. Namely,

$$\text{LTP}_{sct} \;=\; \delta_{FE} \;+\; \tau_1 \cdot \text{OSS}_{sct} \;+\; \tau_2 \cdot \ln(\text{RSC}_{max}) \;+\; \tau_3 \cdot \text{OSS}_{sct} \times \ln(\text{RSC}_{max}) \;+\; \varepsilon_{sct} \tag{3}$$

where RSC_{max} quantifies the distance in shareholder concentrations across companies, defined as the larger of the two ratios, $\max(\text{RSC}_{sct}, \text{RSC}_{cst})$.

Backus et al. (2021) find that RSC (*relative investor concentration* in their paper) explains between one-third and one-half of the total variation in profit weights. Arguably, the apparent rationale is that the variable represents an unbounded ratio with a close-to-Pareto distribution. This feature justifies using the log-ratio of RSC in the estimation, as well as dropping dyads with companies with extremely diffuse ownership, for which RSC_{max} tends to infinity. Thus, I keep observations with RSC within one standard deviation of the mean, which covers roughly 71% of the original sample and 90% of all dyads in which both companies have at least one institutional shareholder or blockholder.

As in the fostering empirical exercise, I normalize LTP, OSS, and RSC to simplify coefficient interpretation, and cluster standard errors at the dyad level.

Altogether, the three fixed-effects specifications complement each other. The first specification—my preferred set of coefficients throughout the analysis—produces esti-

mates that better capture variation across dyads, though it cannot account for potential confounders in the association between customers and suppliers. The other two specifications move in the opposite direction, controlling more flexibly for potential confounders but capturing only within-dyad variation. As such, these specifications likely understate the magnitude of the coefficients, while still supporting the general direction of the effects and interpretations about the role of overlapping shareholders along the supply chain.[12]

B. Using weights to account for infeasible trading partnerships

Trading partnerships represent only a tiny fraction of feasible customer-supplier dyads, suggesting that the sample may still contain many unlikely trading partners. In this subsection, I discuss why infeasible dyads can bias my estimates and how observational weights—proportional to vertical upstream relatedness—can alleviate these concerns.

Unlike previous studies, I avoid conditioning the sample on the prior existence of a trading partnership, allowing me to incorporate the extensive margin into the analysis.[13] That is, studying not only whether the presence of overlapping shareholders affects trading partnerships already in place but also whether they promote new links along the supply chain.

However, populating the sample with every two-firm combination introduces omitted-variable bias, given that whether the two companies can engage in a trading partnership at all remains unobserved. Moreover, restricting the sample to customer-supplier dyads with strictly positive vertical upstream relatedness may not be enough, as the issue still arises, though more subtly.[14]

Let me illustrate the problem and a potential solution with an example. For that, consider a simpler version of Equation (2)

$$\text{LTP}_{sct} = \alpha \;+\; \tau \cdot \text{OSS}_{sct} \;+\; X_{sct} \cdot \beta \;+\; \varepsilon_{sct}$$

in which I replace the fixed effects with some arbitrary set of controls X_{sct}. Suppose the empirical model is correctly specified for feasible trading partnerships, so the WLS estimates for (α, τ, β) that originate from minimizing the following expression

$$\mathbb{E}\Big[\big(\text{LTP}_{sct} = \alpha \;+\; \tau \cdot \text{OSS}_{sct} \;+\; X_{sct} \cdot \beta\big)^2 \omega_{sct}\Big]$$

are unbiased and consistent, where ω_{sct} represent arbitrary observational weights that enter the expression multiplicatively.

In addition, let $\pi\left(F_{sct} = 1\right) \in [0, 1]$ denote the inherent probability that a given trading partnership is feasible, which the econometrician cannot directly observe. When $F_{sct} = 0$, companies cannot engage in trading partnerships that are ultimately infeasible (like the one between Activision Blizzard and Alexion Pharmaceuticals from the

[12] The first and third specifications absorb firm-year error components, so the remaining dependence, if any, lies within the dyad over time, which dyad-level clusters capture appropriately. Conversely, the second specification overlaps with the clustering strategy, meaning that confidence intervals do not account for error dependencies across customers and suppliers and can undercover the true sampling variation.

[13] For example, Fee et al. (2006) and Freeman (2025) focus on supply chain relationships already in place and, as such, can only explore intensive margins, such as sales, or the decision to extend the partnership for an additional year.

[14] One way to think about this limitation is that no company in the sample manufactures a single good or offers a single service, as publicly listed companies in the US tend to be multiproduct firms. Therefore, a supplier needs only to produce at least one commodity the customer uses to produce at least one of its outputs for vertical upstream relatedness to yield a positive but near-zero value. Indeed, the problem spreads as inputs or goods signify a tiny contribution to the whole production process.

example in Subsection I.A), in which case $\text{LTP}_{sct} = 0$. Therefore, I can express the expected error in the estimation as follows

$$\mathbb{E}\Bigg[\mathbb{E}\left[(m_{sct} - f_{sct})^2 \omega_{sct} \middle| F_{sct} = 1\right] \pi\left(F_{sct} = 1\right) + \mathbb{E}\left[(\alpha + \tau \cdot \text{OSS}_{sct} + \beta \cdot X_{sct})^2 \omega_{sct} \middle| F_{sct} = 0\right] \Big(1 - \pi\left(F_{sct} = 1\right)\Big)\Bigg]$$

where m_{sct} and f_{sct} represent the conditional mean and the best predicting function, respectively, when $F_{sct} = 1$ and the customer-supplier dyad is feasible.

Because the estimation minimizes only the first term, failing to account for the feasibility of the customer-supplier dyad can lead to sizable, systematic bias from the second term. Moreover, as $\pi\left(F_{sct} = 1\right)$ approaches zero, the estimator's target becomes irrelevant for predicting the likelihood of a trading partnership.

A first approach would be to model the determinants that make the customer-supplier dyad infeasible explicitly. For instance, by incorporating adequate controls into X_{sct}, it would be possible to offset the bias in the second term without affecting the predictive power from the first term. Unfortunately, such controls are often unavailable, and I must resort to other means.

A second approach is to drop dyads more aggressively when they are suspected to be infeasible, as they systematically contribute to the bias in the second term. Since the highest probabilities of that occurring ensue when $\pi\left(F_{sct} = 1\right)$ is close to zero, setting a higher threshold for vertical upstream relatedness can help alleviate the problem. Moreover, incrementally raising the threshold further reduces concerns, although it can eventually lead me to discard unlikely—though still feasible—dyads from the sample.

A third approach, and the one I follow, is to set weights ω_{sct} that tend to zero when the likelihood of a feasible customer-supplier dyad is low. So far, I have assumed that a null vertical upstream relatedness is informative enough about infeasibility. Therefore, under the same assumption, I can use the same variable as observational weights, downweighting the second term toward zero and significantly reducing bias.

C. Identification strategy: Predicted shareholdings

Investors and managers make portfolio and trading-partnership decisions around the same time, so my empirical strategy requires an exogenous source of variation that also works in vertical settings, where customers and suppliers jointly decide on trading partnerships. This subsection details the steps to construct instruments for OSS and RSC, using predicted rather than observed shareholdings to construct analogous versions of the same metrics.

The literature on common ownership has paid special attention to endogeneity problems associated with using ownership-related measures to explain outcomes at the firm- or market-level (Schmalz, 2018, 2021). Nevertheless, a few examples should suffice to illustrate the same concerns in vertical settings.

Active investors pay close attention to the supply chain before committing to investment decisions, casting doubt on the direction of the mechanisms due to reverse causality. Similarly, imagine a company makes a revolutionary technological innovation. On the one hand, the company may enter new trading partnerships to source the inputs required to produce the new technology. On the other hand, a newly granted patent could attract the attention of several informed investors, increasing their holdings not only in the innovator but also in companies they believe might benefit from the novelty upstream and downstream. This example, along with other unobservable factors, can

simultaneously affect ownership and trading partnerships, leading to omitted-variable bias.

Despite the initial back-and-forth over valid strategies for addressing endogeneity in ownership structures, a few papers document exogenous sources of variation that effectively work around the problem (e.g., Boller and Morton, 2020; Antón et al., 2023; Newham et al., 2025; Azar et al., 2025).[15] However, these studies use strategies whose exclusion restriction relies on the unilateral nature of the decisions they study—such as prices or entry—and do not extend to vertical settings, where customers and suppliers jointly establish trading partnerships. The problem is that index changes shift the ownership structure of at least one company in the dyad and, at the same time, cannot be tied to managerial decisions in other unrelated companies.[16]

The alternative I propose builds on the idea that sufficiently lagged values of a covariate can be valid instruments, provided the outcome's persistence decays faster than the covariate's (Anderson and Hsiao, 1981). Yet my identification strategy takes a turn. Instead of using lagged values of OSS and RSC, I construct analogous versions of these metrics using the predictions from a first-order Markovian process estimated on fifteen years of institutional shareholdings.

This methodology allows me to build instruments for any known function of shareholdings, which is the novel contribution for metrics like RSC. So far, identification strategies relied on shifts in the average OSS across firms and markets, produced by the addition of close competitors into indices and ETFs.

Predicted shareholdings. For the main analysis, I use the following first-order Markovian process for predicting shareholdings

$$\beta_{ijt} \;=\; \alpha \;+\; \rho \cdot \beta_{ijt-1} \;+\; \nu_{ijt} \tag{4}$$

where β_{ijt} denotes the fraction of shares owned by investor $i \in \mathcal{S}$ on company $j \in \mathcal{J}$ in year t.

I argue that the residual variation, ν_{ijt}, captures most of the endogeneity that biases OLS estimates, assuming the chosen number of lags satisfies the exclusion restriction. In other words, as long as the supply chain persistence decays faster than institutional shareholdings persistence. Subsection C.3 in the Appendix reports a robustness analysis and shows that alternative specifications of the prediction model do not change the paper's conclusions. If anything, using fewer lags stands as the conservative choice.

Given that the model uses one-year lagged shareholdings, the estimation automatically excludes institutional owners in years when they first purchased shares in a company. This becomes a byproduct of the instrumental variables approach, but contributes to the interpretation that compliers belong to the portfolio of shareholders with fewer perturbations in their holdings—as these companies will tend to have a stronger first stage.[17] Subsection C.4 in the Appendix extrapolates the intuition to companies with more stable ownership structures. After excluding companies in years

[15] For the initial debate and exchange of opinions, see Kennedy et al. (2017), Azar et al. (2018), Dennis et al. (2022), and Azar et al. (2022). Other examples include critiques in Berger (2023), who describes problems associated with fire sales resulting from mutual fund scandals, and Lewellen and Lowry (2021), w ho discusses the limitations of financial institution mergers and additions to indexes.

[16] Several papers use index changes as exogenous sources of variation. Boone and White (2015), Kennedy et al. (2017), Brooks et al. (2018), and Kostovetsky and Manconi (2020) use Russell Index reconstitutions, while Aghion et al. (2013) use S&P 500 additions. The novelty in Boller and Morton (2020) lies in not considering the outcomes of the added companies themselves, but those of competitors, for whom the changes in index constituents are completely unrelated.

[17] Because both covariates and their analogous instruments are continuous, the 2SLS estimand (LATE) corresponds to the average effects across compliers, weighted proportionally by its first-stage marginal effect (Angrist et al., 2000).

when they participated in, or were related to, S&P 500 index changes, I find no statistically significant differences in estimates.

I use OLS to estimate Equation (4) on the same ownership network described in Section I. The ownership network comprises panel data on annual ownership ties between US companies and institutional shareholders or blockholders, spanning 1999 to 2013. The estimation yields the following coefficients

$$\underset{}{\alpha} \quad \underset{(0.000010)}{0.0005} \qquad \underset{}{\rho} \quad \underset{(0.002219)}{0.9075} \qquad R^2 \quad 0.814$$

all of which are statistically significant at the 99% confidence level when using standard errors robust to heteroskedasticity. More importantly, the R^2 is 0.814, suggesting that past shareholdings explain a sizable portion of the variation in portfolios.

Predicted and observed ownership structures differ, as estimates from Equation (4) do not fully capture sudden portfolio shifts. Thus, my approach treats these holdings as if they were in the hands of retail shareholders—individual investors who buy and sell company shares in small amounts—when computing ownership structures.

I feed the predicted ownership structures back into the formulas described in Subsection I.C. This procedure yields analogous versions of OSS and RSC that I use as instruments, muting the endogenous response of investment decisions without sacrificing the non-linearities embedded in the overlapping ownership metrics. Furthermore, I build similar instruments for a few alternative overlapping ownership metrics in the literature, which I cover in detail in Subsection C.2 in the Appendix.

As discussed above, the exclusion restriction entails a crucial identification assumption. As such, fixed effects must effectively absorb supply chain persistence in both steps of the 2SLS estimation to keep estimates unbiased. The first and third specifications—which include time-varying customer and supplier fixed effects—are more likely to capture company-specific trends and control for persistence driven by specific customers and suppliers. Likewise, the second and third specifications—which include dyad fixed effects—control for supply chain persistence driven by specific trading partners. Together, these considerations reinforce the idea that all three specifications complement each other in interpreting the results.

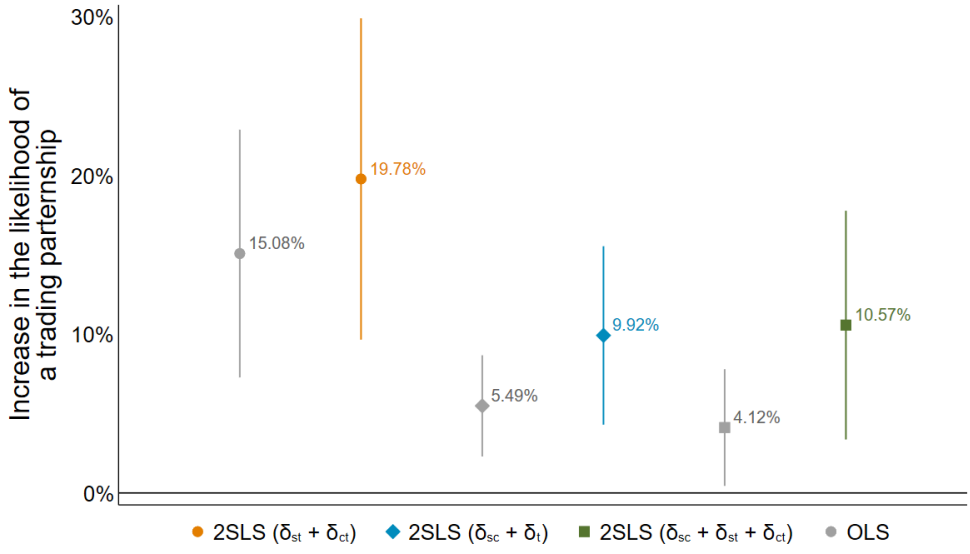
A potential argument against my instrument is that supply chain persistence depends on idiosyncratic, time-varying characteristics of the trading partnership itself. Unfortunately, such dependence coincides with the identifying variation and is therefore beyond the scope of my empirical strategy.

III. Overlapping shareholders: Turning vertical relations into private benefits?

Whereas overlapping shareholders may foster trading partnerships among companies in their portfolios, the gains from these vertical relations need not be shared equally, turning many trading partnerships into one-sided vertical relations over time. The following section offers suggestive evidence that institutional shareholders promote trading partnerships that create private benefits from tunneling opportunities. Specifically, they influence managerial decisions to transfer profits and equity value toward companies with concentrated ownership, at the expense of non-overlapping shareholders in companies with diffuse ownership.

My results align with previous literature, which shows evidence that OSS between customers and suppliers increases the likelihood of active trading partnerships. While prior studies argue that overlapping shareholders foster trading partnerships by reducing contractual and informational frictions—such as holdup (Deng et al., 2025; Freeman, 2025) and double marginalization (Riva, 2022)—this paper focuses on a different mech-

Figure 1 – *Overlapping ownership increases the likelihood of active trading partnerships*



anism. Nevertheless, Section B in the Appendix presents additional empirical exercises associated with these mechanisms. Overall, the same identification strategy I use to identify fostering and tunneling effects yields empirical patterns consistent with the literature.

A. Overlapping shareholders foster trading partnerships

This subsection presents empirical evidence that the likelihood of an active trading partnership increases with the OSS between customers and suppliers, suggesting that overlapping shareholders foster trading partnerships among companies in their portfolios.

For this empirical exercise, I estimate Equation (2) using OLS and 2SLS, as described in Section II. Two distinct sources of bias can distort OLS estimates for OSS. On the one hand, measurement error in ownership structures—likely due to unobserved insider and retail holdings—creates attenuation bias that pushes coefficients toward zero. On the other hand, omitted-variable bias can shift estimates in either direction, depending on whether unaccounted determinants of trading partnerships positively or negatively correlate with ownership.

Figure 1 reports the coefficients across specifications, which indicate that a one-standard-deviation increase in OSS raises the average probability of a trading partnership by 9.92% to 19.78%, depending on the fixed-effects specification. For details, see Table SA.1 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

The three specifications convey a single message: overlapping shareholders in feasible customer-supplier dyads foster trading partnerships, despite differences in magnitude. For instance, my preferred specification (orange) points to around a 20% increase in the average probability, while the ones that include dyad fixed effects (blue and green) settle for roughly a 10% increase. Considering that the later specifications exploit only within-dyad variation, the combined results strengthen the case that overlapping ownership fosters trading partnerships rather than merely reflecting between-dyad selection.

Notably, coefficients are slightly higher in the 2SLS estimation, despite being sta-

tistically indistinguishable from their OLS counterparts. Freeman (2025) reaches the same conclusion using financial institutional mergers as their main identification strategy. Thus, 2SLS estimates either alleviate a seemingly modest attenuation bias or a positive bias of comparable magnitude (e.g., reverse causality) with the former offsets it.

I repeat the empirical exercise using a second-order Markovian process for predicting ownership structures. The comparison across the two prediction models implies a negative correlation between ownership and the unaccounted determinants of trading partnerships, meaning the instrumental variables approach accounts for combined, though modest, attenuation and omitted-variable bias. For a comparison of the results across prediction models, see Subsection C.3 in the Appendix.

Building on this interpretation, 2SLS estimates report local average treatment effects (LATE), and complier dyads in my setting involve companies with more stable ownership structures over time—such as passive index investors and institutional shareholders with long-term stakes. Thus, the similarity between OLS and 2SLS estimates may also shed light on the mechanism, emphasizing which shareholders are most likely to influence managerial decisions.

Arguably, long-standing investors in companies have the most influence over managers: not necessarily because they directly persuade executives, but because managers tend to cater to pivotal shareholders with longer investment horizons. Thus, managers weigh more heavily how their choices affect other companies in those investors' portfolios.

So far, the findings establish that overlapping shareholders encourage trading partnerships among companies in their portfolios; whether those business relationships yield symmetric gains for customers and suppliers is a separate question. The next subsection presents suggestive evidence that the same ownership overlap can create asymmetric managerial incentives, leading to tunneling opportunities along the supply chain.

B. Tunneling: Privately benefiting from trading partnerships

This subsection focuses on a mechanism inspired by certain features of partial vertical integration: overlapping shareholders use the supply chain for their private benefit. In other words, managers make decisions about trading partnerships that negatively affect one company and, in particular, its non-overlapping shareholders.

But, as with partial vertical integration, the exact mechanism can take many forms. For instance, managers could foreclose upstream or downstream markets to companies outside shareholders' portfolios, create tighter market-entry barriers, or directly tunnel equity value or profits from one company to another, as when firms set different *transfer prices* to shift profits from one division to another.

This paper presents suggestive evidence that companies prefer trading partnerships with asymmetric managerial incentives, although it cannot pin down the exact mechanism or channel. The baseline effect of increasing OSS roughly quadruples in customer-supplier dyads where a company has a significantly more diffuse ownership. In what remains of the section, I argue that the observed patterns reflect managers weighing gains with trading partners more heavily than losses at the companies that appointed them.

Tunneling. The first mechanism that comes to mind when overlapping shareholders privately benefit from managerial decisions at the expense of other investors is *tunneling* (Johnson et al., 2000; Atanasov et al., 2007). That is, when a company transfers assets or cash flows to another. For example, consider a group of overlapping shareholders who own 60% of the equity in customer c and 20% of the equity in supplier s, and suppose these shareholders can prompt managers to set a lower input price than

otherwise would be the case. Then, for every USD 100 million the customer saves on input costs, overlapping shareholders would pocket a net difference of USD 40 million. Furthermore, non-overlapping shareholders in the downstream company would benefit as well, obtaining another USD 40 million, while non-overlapping shareholders in the upstream company would lose the entire USD 80 million.

However, as Ehrhardt and Nowak (2015) explains, tunneling takes many forms and does not necessarily imply monetary transfers. Overlapping shareholders may pursue other goals, such as extending their control over firms, the supply chain, or the degree of competition across markets.

As such, tunneling is analogous to many of the mechanisms associated with partial vertical integration, which the literature has documented and discussed as a separate issue. For example, Boehm and Sonntag (2023) finds evidence that vertical mergers increase the likelihood of foreclosure, and Newham et al. (2025) show that pharmaceuticals can deter the entry of generics. Similarly, Levy et al. (2018) develop a theoretical framework in which companies can acquire partial stakes in vertically related companies to foreclose rivals, which can be profitable, depending on the ownership and control structures of the parties involved.

The framework in Rotemberg (1984) offers a broad interpretation that connects the implications of tunneling, partial vertical integration, and overlapping ownership. The model posits that managers assign a positive profit weight to a company with whom ownership overlaps, with Backus et al. (2021) going so far as to label profit weights above one as tunneling, given that managers would strictly prioritize the weighted firm over the one that appointed them. Nevertheless, smaller profit weights affect managerial decisions, especially in vertical settings, where asymmetric incentives can shift how companies are willing to split the gains from doing business together.

All in all, overlapping ownership has implications similar to those of partially integrated companies, which can, in turn, create tunneling opportunities—in a broad sense. That is, not only tunneling as transfers of equity value and profits across companies, but also as decisions that leave overlapping shareholders better off than they would otherwise be.

For this empirical exercise, I estimate Equation (3) using OLS and 2SLS, as described in Section II. For details, see Table SA.2 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages.

As discussed in Subsection I.C, RSC is a scale-only statistic, so OSS is the sole component that contains information about the shape of profit weights. In other words, according to the framework I use to derive these metrics, RSC should affect the outcome only through OSS. Therefore, when reporting the results, I find it more transparent to convey their interaction using the following expression for the heterogeneous marginal effects

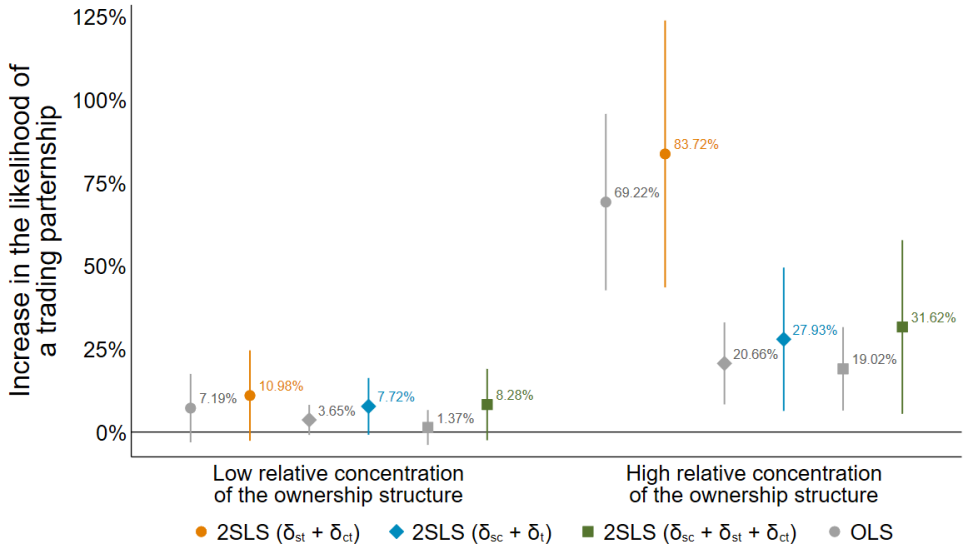
$$\tau(\text{RSC}) \;=\; \tau_1 \;+\; \tau_3 \cdot \ln(\text{RSC}_{max})$$

rather than the interaction coefficient alone. Thus, $\tau(\text{RSC})$ reports the marginal effect of a one-standard-deviation increase in OSS, holding the distance in shareholder concentration across companies, $\ln(\text{RSC}_{max})$, constant.

Figure 2 displays the marginal effects across estimations. Coefficients on the left report marginal effects at the bottom decile of the RSC_{max} distribution (1.0889), while coefficients on the right do so at the top decile (5.2089). In this way, the figure separates the margin on which managers decide to attend to a trading partner's profits at all (bottom decile, nearly identical concentration) from the margin on which that attention becomes asymmetric (top decile, strong asymmetric managerial incentives).

Coefficients on the left indicate that an increase in OSS has no statistically significant

Figure 2 – *More diffuse shareholdings amplify the effects of increasing the degree of overlapping ownership*



effect on the likelihood of a trading partnership when RSCs are close to one. That is, when managers in the customer-supplier dyad assign almost identical profit weights to each other's companies. By contrast, the coefficients on the right indicate that a one-standard-deviation increase in OSS raises the average probability of a trading partnership by 27.93% to 83.72%, depending on the fixed-effects specification, roughly quadrupling fostering estimates.

The combined results offer an alternative reading of results in other empirical papers, as ownership overlap does not appear to drive the results in Subsection III.A on its own. Studies that use single composite metrics, such as *overlapping institutional ownership* (Gao et al., 2022; Deng et al., 2025; Freeman, 2025), conflate the shape and scale margins of overlapping ownership. Therefore, the mechanisms in those studies may respond to the asymmetric-concentration margin rather than overlapping shareholders alone.

The comparison, however, supports the interpretation that managers are more inclined to establish and extend one-sided trading partnerships over time. Moreover, even if the empirical exercise remains silent on the exact channel, the framework underlying OSS and RSC metrics offers clear intuition.

Intuitively, I argue that trading partnerships become more likely when managers weigh the externalities imposed on the potential trading partner more heavily. In particular, when the trading partner's manager does not reciprocate. These decisions enable tunneling opportunities from companies with diffuse ownership to those with concentrated ownership. Overlapping shareholders can then privately benefit from these opportunities and, in the process, harm non-overlapping shareholders at the diffuse-ownership firm that appointed the conflicted manager.

SFAS N°131—Censoring as a lower bound. A natural concern is that marginal effects may reflect the censoring introduced by SFAS N°131 rather than a genuine economic mechanism. As discussed in Subsection I.A, the Statement requires disclosure only when a customer accounts for at least 10% of a supplier's annual sales. Therefore, small customers and large suppliers are mechanically less likely to appear in the Customer Segment.

To test this further, I split the sample by the side of the trading partnership with the more concentrated ownership structure. If censoring were driving the results, one would expect significant differences across subsamples. First, trading partnerships with diffuse-ownership suppliers would exhibit smaller sales ratios to meet the SFAS N°131 threshold. Second, disclosure rates would differ across subsamples. Third, customer and supplier shareholder concentrations would drastically differ across subsamples.

The comparison shows none of these patterns. First, sales ratios on trading partnerships are roughly 100 times larger than on other customer-supplier dyads, but are around the same order of magnitude across subsamples. Second, disclosure rates are nearly identical across subsamples, with around 77

The descriptives confirm censoring, but they do not support it as the source of asymmetric managerial incentives. Moreover, the two specifications that include dyad fixed effects should absorb most of the remaining association between the threshold and RSC.

Nevertheless, unobserved trading partnerships between large suppliers and small customers can still bias my estimates toward zero. Because the sales ratio in customer-supplier dyads likely correlates with RSC, censoring disproportionately zeros out trading partnerships where the supplier has diffuse ownership, and the customer is concentrated—the very dyads where the empirical model predicts tunneling effects to be strongest.

Therefore, I estimate Equation (3) separately on each subsample. For details, see Table SA.3 in the Appendix, which reports OLS and 2SLS coefficients and statistics, including the corresponding first stages. Notice that by splitting the sample, time-varying company fixed effects fully absorb RSC, so it is dropped from the estimation.

Figure 3 supports the intuition but also suggests caution when speaking to the magnitudes of these effects. For instance, Panel (a) reports coefficients solely for dyads where customers have more diffuse ownership. While the conclusions align with those of Figure 2, the magnitudes under asymmetric managerial incentives are systematically larger, suggesting that earlier results may constitute a lower bound on the true effects. The estimates report that a one-standard-deviation increase in OSS raises the average probability of a trading partnership by 89.32% to 164.10%, depending on the fixed-effects specification.

In contrast, Panel (b) reports coefficients only for dyads where suppliers have more diffuse ownership, which are harder to interpret individually because of censoring. Taken at face value, the point estimates suggest that RSC reduces the effect of increasing OSS when customers are more concentrated. However, as the outcome in this subsample contains much more noise, confidence intervals are larger, and coefficients are not statistically distinguishable from zero.

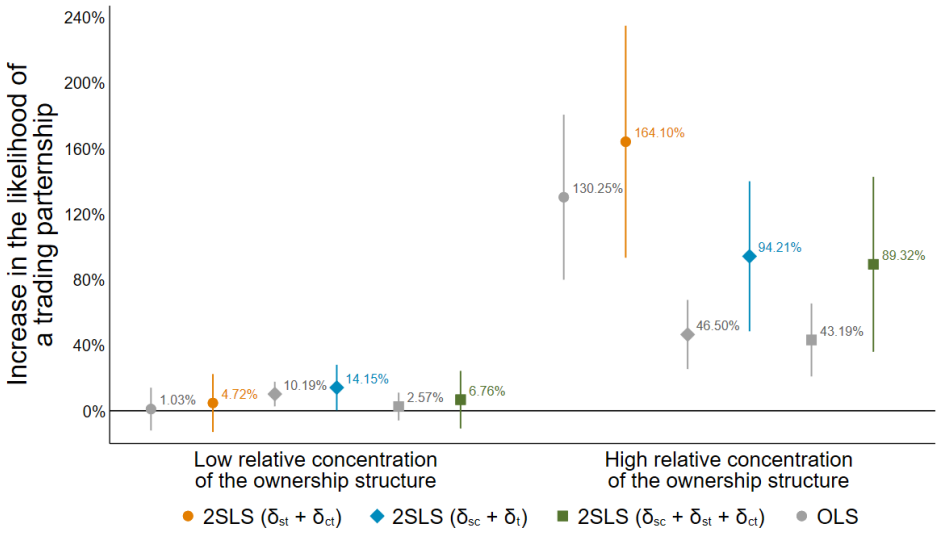
In conclusion, the asymmetric-concentration margin appears larger than the previous empirical exercise suggested. However, the unobserved trading partnerships in the data coincide with the very dyads where expected tunneling effects are strongest. Therefore, at this point, it is impossible to anticipate where the diffusion mechanism is truly symmetric or whether it responds to economic mechanisms solely on the customer side of the relationship.

IV. Conclusion

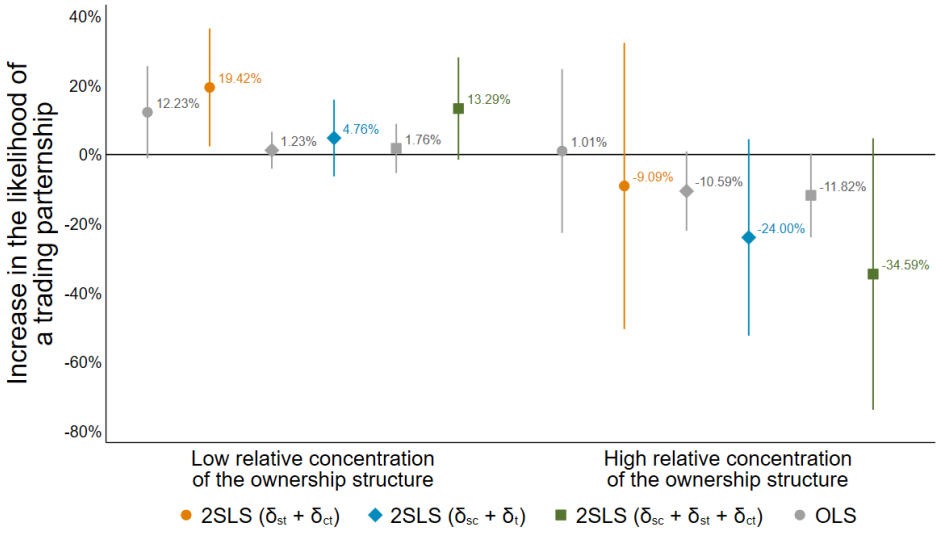
This paper contributes to the growing literature on common ownership. While early work has shown that overlapping shareholders can influence managerial decisions to soften competition among rivals (e.g., Azar et al., 2018; Torshizi and Clapp, 2021; Newham et al., 2025), I ask whether the same mechanism operates when companies lie on opposite sides of a potential trading partnership.

Figure 3 – *Tunneling effects are larger but concentrated among dyads with diffuse-ownership suppliers*

(a) Diffused-ownership customer



(b) Diffused-ownership supplier



Recent studies on overlapping ownership in vertical relations emphasize the role of institutional shareholders in mitigating contractual and informational frictions (Freeman, 2025; Deng et al., 2025), which support the idea that overlapping ownership and partial vertical integration share several features and mechanisms.

My work takes a different route and focuses on mechanisms analogous to tunneling (Johnson et al., 2000). By combining several open data sources from 1999 to 2013 on US ownership and supply chain networks, I find that overlapping shareholders in customer-supplier dyads increase the likelihood of trading partnerships. Interestingly, this effect strengthens when these shareholders can extract private benefits from the supply chain, multiplying the baseline effect by a factor of four. In other words, the effect strengthens when managers appointed by companies with diffuse ownership become more inclined to transfer resources out of the company toward other potential trading partners with more concentrated shareholdings. These results also hold across alternative sample definitions, ownership metrics, and prediction models associated with the instrumental variables approach.

The analysis employs a novel identification strategy that goes beyond unilateral decisions and aggregate shifts in overlapping ownership at the market- or firm-level (Boller and Morton, 2020; Antón et al., 2023; Azar et al., 2025). I first predict ownership structures using a first-order Markovian process on fifteen years of panel data on US shareholdings. I then feed the predictions into the same formulas that yield the overlapping ownership metrics I use throughout the paper, creating analogous versions that serve as instruments.

My empirical strategy distinguishes two margins associated with overlapping ownership. First, the overlap itself, which captures whether managers attend to each other's profits at all. Second, the margin on which managerial attention becomes asymmetric. I present suggestive evidence that the latter dominates, meaning that trading partnerships become more likely when executives weigh more heavily the externalities they impose on the trading partner than the effects on the firm that appointed them in the first place. Furthermore, the channel does not require overlapping shareholders to exert direct influence over executives; it may be a byproduct of their mere presence on both sides of the trading partnership.

Decisions to establish or sustain such trading partnerships enable tunneling opportunities from companies with diffuse ownership to those with concentrated ownership, which can ultimately leave non-overlapping shareholders in those companies worse off. This finding reintroduces the classical tension between ownership and control, shifting the axis of conflict from managers versus owners to overlapping versus non-overlapping shareholders.

Taken together, my findings challenge the interpretation of many results in the existing empirical literature. Prior studies have used composite metrics that conflate the shape and scale margins of overlapping ownership. As a result, many of the mechanisms in those studies may respond to the asymmetric-concentration margin rather than overlapping shareholders per se.

This paper also indicates promising directions for future research. For instance, it remains unclear whether conflicted managers engage in cash-flow tunneling, or whether overlapping shareholders prioritize other goals. The literature on partial vertical integration suggests a few mechanisms worth exploring, e.g., foreclosing companies outside their portfolios, deterring new entrants, or gaining further control over the company, in the supply chain, or across markets.

Furthermore, the empirical exercises cannot pin down the exact channels in play. This leaves open whether non-overlapping shareholders suffer direct losses or foregone profits—especially because trading partnerships tend to create value for the parties

involved. Even if overlapping shareholders capture a significant share of the value generated as private benefits, companies with diffuse ownership may still reap some of the gains.

Finally, it remains to be seen whether the mechanisms identified in the literature apply when trading partners have asymmetric managerial incentives, or whether this interpretation may yield new channels. In particular, mechanisms associated with the disclosure of private information (Boone and White, 2015; Pawliczek et al., 2022), say-on-pay (Fichtner et al., 2017; Bubb and Catan, 2022; Antón et al., 2023), and interlocking boards (Azar, 2022) seem better suited when overlapping shareholders aim for coordination rather than leveraging a one-sided relationship. Other promising avenues include managers on the payroll of institutional funds (Freeman, 2025), selective omission (Wells, 2016; Hemphill and Marcel, 2019), and, especially, tying the compensation of executives to ESG-based outcomes (Bebchuk and Tallarita, 2022; Cohen et al., 2023), which can restrict customers or suppliers to a selected set of companies that meet very specific characteristics.

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APPENDIX

A. Framework: Derivation of profit weights

This appendix depicts the steps to derive profit weights from a framework first introduced by Rotemberg (1984). For a more complete description of the model's assumptions and properties, see Backus et al. (2021).

Shareholders have no particular interest in any single company in their portfolios. From their standpoint, the goal is to maximize cash flow rights across all companies in which they hold shares, which amounts to the overall market value of their portfolios. Hence, let V_{it} represent the total portfolio value of investor $i \in \mathcal{S}$ during year t, which can be written as

$$V_{it}(x_{jt}, x_{-jt}; \beta_t) = \sum_{\forall j \in \mathcal{J}} \beta^i_{jt} \pi_{jt}(x_{jt}, x_{-jt}) \tag{A.1}$$

where β^i_{jt} designates the fraction of outstanding shares that investor i holds in the company $j \in \mathcal{J}$ in year t. In addition, π_{jt} depicts the profits accrued by company j in year t, which jointly depends on an arbitrary set of managerial decisions, x_{jt} and x_{-jt}. The former represents decisions made by executives appointed by company j, while the latter represents decisions taken by executives appointed in all other companies in the economy.

The author posits that, contrary to what is typically assumed in the literature, shareholders appoint managers not to maximize company profits but to pursue shareholder welfare (Rotemberg, 1984). This claim resonates with several statements in the corporate finance and law literature: that a firm should always answer to its investors (Jensen and Meckling, 1976), and it also anticipates a shift in power dynamics between shareholders and managers that has occurred since the late 1990s. For instance, Wells (2016) lists a series of events that contributed to the rise of shareholder power and the crystallization of two critical instruments for stockholder activism: jawboning and shareholder proposals.[18]

The first began in 1992, when the SEC revised the proxy solicitation rules. While the original goal was to ease communication between large shareholders, they inadvertently promoted the proliferation of meetings and other less formal, less visible interactions between shareholders and corporate management. The second became widely used in the mid-1990s, leading to the formation of several shareholder coalitions that discuss and draft proposals daily.[19]

Thus, the framework proposes a setting in which managers maximize a weighted average of the overall market value of shareholders' portfolios. The objective function of managers builds on the work of Drèze (1974) and Grossman and Hart (1979), so that managers maximize the following expression

$$Q_{jt}\left(x_{jt}, x_{-jt}; \gamma_t\right) = \sum_{\forall i \in \mathcal{S}} \gamma^i_{jt} \; V_{it}(x_{jt}, x_{-jt}) \tag{A.2}$$

where γ^i_{jt} measures the control that investor i exercises in company j in year t. There-

[18] For example, the author mentions the creation of the Institutional Shareholder Services (ISS) in 1985, the issuance of the *Avon Letter* by the US Department of Labor in 1988, the enactment of the Dodd-Frank Act by the US Congress in 2010, and the closer ties between unions and institutional shareholders—fostered by an increasing focus on pension funds since the mid-1990s. For a more in-depth explanation of how these changes affected the corporate power dynamics, see Wells (2016).

[19] For example, James McRitchie, founder of the blog Corporate Governance, acted as the plaintiff in a lawsuit against Meta Platforms Inc. executive officers, including its CEO, Mark Zuckerberg. The accusation is that board members own an excessive shareholding in the company, thereby making decisions that, though beneficial for Meta, ignore their effects on stockholders' portfolios. This is the exact mechanism proposed in Rotemberg (1984). For the full version of the document, refer to this link.

fore, when managers seek to maximize shareholder welfare through their actions, x_{jt}, they use γ_{jt}^i to reflect the relative importance of each specific portfolio.

Combining Equations (A.1) and (A.2) yields an expression for the profit weights managers assign to other companies in the economy, which in turn constrain the decisions that managers can take when those decisions affect other firms' outcomes.

$$\begin{aligned} Q_{jt}\left(x_t;\beta_t,\gamma_t\right) &= \sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \;\; V_{it}(x_t) \\ &= \sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \sum_{\forall k\in\mathcal{J}} \beta_{kt}^i \pi_{kt}(x_t) \\ &= \sum_{\forall k\in\mathcal{J}} \sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \;\; \beta_{kt}^i \;\; \pi_{kt}(x_t) \\ &\propto \pi_{jt}(x_t) + \sum_{k\in\mathcal{J}-\{j\}} \left(\frac{\sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \;\; \beta_{kt}^i}{\sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \;\; \beta_{jt}^i} \right) \pi_{kt}(x_t) \\ &\propto \pi_{jt}(x_t) + \sum_{k\in\mathcal{J}-\{j\}} \kappa_{jkt} \;\; \pi_{kt}(x_t) \end{aligned}$$

where x_t denotes both the endogenous vector of managerial decisions x_{jt}—for executives appointed to company j—and the corresponding set of response functions for companies in $\mathcal{J}-\{j\}$: x_{-jt}.[20] The interpretation of κ_{jkt} is straightforward. Profit weights determine how the manager appointed by company j weights the profits of company k in year t when making an arbitrary set of decisions x_{jt} that impose externalities on that company's profits.

Furthermore, using vector notation, I can rewrite the expression for profit weights as a ratio of two inner products. Interestingly, this transformation into polar notation yields a decomposition that clearly illustrates the underlying mechanisms.

$$\begin{aligned} \kappa_{jkt} &= \frac{\sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \;\; \beta_{kt}^i}{\sum_{\forall i\in\mathcal{S}} \gamma_{jt}^i \;\; \beta_{jt}^i} \\ &= \frac{\langle \gamma_{jt}, \beta_{kt} \rangle}{\langle \gamma_{jt}, \beta_{jt} \rangle} \\ &= \frac{cos(\gamma_{jt}, \beta_{kt}) \cancel{\|\gamma_{jt}\|} \|\beta_{kt}\|}{cos(\gamma_{jt}, \beta_{jt}) \cancel{\|\gamma_{jt}\|} \|\beta_{jt}\|} \end{aligned}$$

The first factor is a shape-only statistic that compares the overlap between control and cash flow rights across companies. The denominator tells the manager in company j how similar control and cash flow rights are within the same firm that appointed them. The numerator, instead, uncovers the overlap between control rights in company j and cash flow rights in company k. Therefore, a higher magnitude on the first factor indicates that managers at both company j and company k pay attention to each other's profits.

The second factor is a scale-only statistic that represents ownership concentration (in terms of cash flow rights) across the two companies. So, a higher magnitude on the second factor has two implications. First, the manager in company j pays closer attention to how their decisions affect company k. Second, company k becomes more

[20] The fourth step in the derivation stems from a simple normalization, so that every manager assigns a profit weight equal to 1 to the company that appointed them.

self-centered, and its manager pays less attention to the effects on company j. Therefore, this metric captures not only the intensity but the direction of the attention in managerial decisions.

Note that using profit weights requires appropriately distinguishing between ownership and control. In this paper, I assume proportional control, implying that γ_{jt} equals β_{jt}.[21] As a result, profit weight can be expressed as

$$\kappa_{jkt} = cos(\beta_{jt}, \beta_{kt}) \frac{\|\beta_{kt}\|}{\|\beta_{jt}\|}$$

B. Mechanisms associated with partial vertical integration (WIP)

The literature on industrial organization has shown that vertical integrations can affect upstream and downstream market outcomes (Bolton and Whinston, 1993; Boehm and Sonntag, 2023). Partial integration and vertical overlapping owners can attain similar results, but without companies engaging in mergers and acquisitions. For example, Fee et al. (2006) explore the role of partial vertical integration on the stability of customer-supplier relationships, while Freeman (2025) extrapolates similar results to third-party overlapping shareholders. Therefore, it seems reasonable to consider that overlapping shareholders affect the supply chain network through typical features of vertical integration and vertical control.[22]

To begin with, overlapping owners could enable valuable trading partnerships that otherwise would not exist due to contractual frictions, such as incomplete contracts or injunctions from competition policy authorities.[23] For example, by easing access to data, overlapping shareholders could help to reduce the cost of getting information about prospective trading partners and alleviate conflicting interests. Since the screening process and transaction costs occur on both sides of the supply chain relationship, they could make it more likely for companies with similar ownership structures to pick each other. Moreover, competition policy authorities often overlook ownership ties across companies, so overlapping shareholders could act as makeshifts for the contracts, mergers, and acquisitions that authorities would deem anti-competitive behavior. Similarly, they could help work around the discouraging costs associated with such financial transactions.

Beyond nudges at the customer-supplier dyad, overlapping shareholders could make systemic risk more evident for individual companies. Therefore, it could be that managers internalize the benefits of engaging with redundant trading partners to reduce the impact of natural disasters (Carvalho et al., 2021; Barrot and Sauvagnat, 2016), bullwhip effects (Croson and Donohue, 2005), wars, or other external shocks (Wang, 2025). Moreover, they could increase supply reliability for customers (Bolton and Whinston, 1993; Wu and Birge, 2014), alleviate suppliers' cash constraints (Fee et al., 2006), smooth sales dependence on inherently uncertain markets, or provide companies that hold or require essential facilities with a competitive advantage.

[21] The assumption is not harmless, but it has been standard in the literature for many years. A few papers, however, have pursued a different direction. For example, Gilje et al. (2020) derives an alternative metric that incorporates attentiveness through a corporate voting model, while Antón et al. (2025*b*) represents control rights as a quadratic form of cash flow rights. For a discussion of the tradeoffs between cash flow and control rights—incentives versus influence in tunneling—see Newham et al. (2025).

[22] Vertical integration is a way of organizing a trading relationship in which a company aims to gain control of multiple steps along the supply chain. Instead, vertical control implies transferring decision-making rights of some but not all aspects of the trading partnership.

[23] For instance, revenue-sharing contracts have been under scrutiny by competition policy authorities in both the US and Europe. As an example, the Directorate-General for Competition of the European Commission accused Google in 2018 of using these contracts to bundle Google Chrome and the PlayStore to the Android OS.

Subsection III.B explores an alternative mechanism, emphasizing the role of tunneling. Nonetheless, transferring value across companies may take many forms and does not necessarily entail cash flows. In particular, overlapping shareholders could seek tighter control over the supply chain, thereby gaining an advantage across multiple markets simultaneously.

This interpretation resembles the typical drawbacks of partial vertical integration in the literature. For example, a higher ownership-structure similarity could lead to foreclosure in upstream and downstream markets (Levy et al., 2018; Boehm and Sonntag, 2023) or tighter market-entry barriers (Newham et al., 2025). I leave Subsection III.B for the analysis of such mechanisms, focusing here on the positive aspects discussed in prior studies: holdup and double marginalization.

Since these particular mechanisms fall beyond the scope of the paper, I offer only suggestive evidence that aligns with prior studies. These investigations leverage better settings and specific sources of variation, demonstrating the validity of the particular channels. Nevertheless, the relative importance of each mechanism remains to be determined.

Holdup. Customers and suppliers often face the opportunity to boost profits through coordination and long-term contracting. However, contractual frictions may preclude potential trading partners from pursuing these opportunities. Alternatively, these companies can opt to merge or acquire each other as a workaround.

However, the costs associated with merging or acquiring each other can be high, so companies would pursue these strategies only when signing a contract is not an option. Nonetheless, contracting may become completely unfeasible under certain circumstances. Either due to uncertainty or information asymmetries, a few situations can arise that prevent companies from agreeing on the contract's content, on who should retain residual control over it, or how to substitute specific clauses that competition policy authorities often deem as anticompetitive.

Holdup constitutes a typical example of contractual frictions between customers and suppliers. The core of the problem lies in the disagreement over how to split profits originating from partnership-specific investments. In particular, because agents cannot truthfully commit without a binding document, one of the trading partners refrains from making the investment, concerned about being held up by the other party.

The literature presumes that overlapping shareholders can help alleviate these informational problems, either by disclosing otherwise private information (Pawliczek et al., 2022; Boone and White, 2015) or by enforcing tacit agreements through other means, including voting (Fichtner et al., 2017; Bubb and Catan, 2022; Antón et al., 2023), interlocking boards (Azar, 2022), putting managers on the payroll of institutional funds (Freeman, 2025), tying compensation to ESG-based outcomes (Bebchuk and Tallarita, 2022; Cohen et al., 2023), or through selective omission (Wells, 2016; Hemphill and Marcel, 2019).[24]

Along the same lines, prior studies have found empirical support for this intuition. For instance, Freeman (2025) shows that overlapping shareholders lengthen trading partnerships while improving innovation outcomes, and Deng et al. (2025) finds that suppliers invest more intensively in partnership-specific assets. I further examine the possibility by exploring the heterogeneous effects of overlapping ownership across mar-

[24] *Interlocking boards* refers to institutional shareholders who have a seat on the board of multiple firms. *ESG-based compensation* ties executive pay to environmental, social, and governance metrics, which can restrict customers or suppliers to companies meeting very specific characteristics. *Selective omission* represents situations in which shareholders encourage actions that simultaneously benefit the company's profits and investors' portfolios, but remain silent otherwise. Wells (2016) provides a few examples that illustrate how institutional investment funds tipped the scales and amplified their voice in corporate decision-making through several means at their disposal, e.g., jawboning, shareholders' proposals, proxy voting, among others.

kets where holdup appears to be more prevalent.

The literature broadly considers three dimensions to identify innovation activities related to holdup (Aghion et al., 2013; Freeman, 2025; Deng et al., 2025; Antón et al., 2025*a*). The innovation input reflects the resources and efforts invested by firms; the innovation output represents the outcomes of the innovation process; and the innovation specificity refers to the degree of customization to a particular trading partner's needs. I measure the prevalence of holdup by classifying companies by their level of R&D intensity. That is, the ratio of annual R&D expenses to total book assets, which is a standard proxy for innovation input in the literature. A caveat of using only innovation input is that the variable captures only the average likelihood that companies may face holdup problems with any given trading partner, unlike input specificity, which provides pairwise measures that would allow a classification of customer-supplier dyads rather than individual companies or market aggregates.

I classify customers and suppliers based on whether they are above or below the average R&D intensity (6.89%) among firms in the sample, i.e., publicly listed US companies between 1999 and 2013, as described in Section I. I then extend Equation (X)

Resolved 20/10/2026: the clause here now points to Section I, where the full screen is stated once (the pipeline drops the financial, transportation, communication and utility sectors), and the same substitution was applied to the double-marginalization classification below and to the notes of the 14 estimation exhibits. Table 1's note is the deliberate exception—it describes Panel A, so naming the industries is the description of the panel.

$$Link_{sct} = \delta_{st} + \delta_{ct} + OvrOwn_{sct} \times \sum_{j\in\{L,H\}} \sum_{k\in\{L,H\}} \tau^{jk} \, (R\&D)^{jk} + \varepsilon_{sct} \tag{B.1}$$

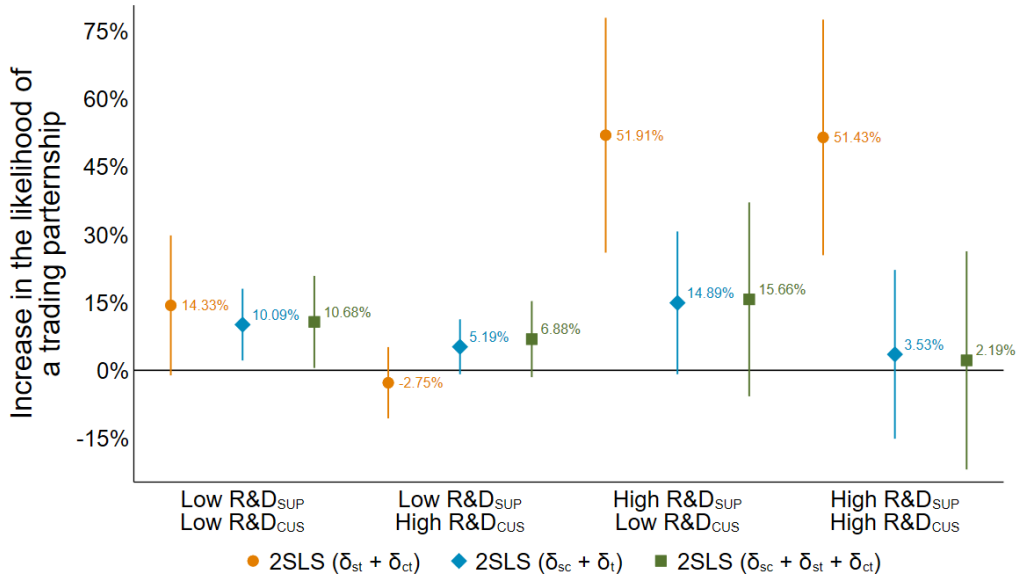
where $(R\&D)^{LH}$ indicates that customer c has a relative low R&D intensity, while supplier s has a relatively high intensity. The same interpretation applies to the remaining dummies and the four coefficients τ^{LL}, τ^{LH}, τ^{HL}, and τ^{HH}. Likewise, I report results from extending Equation (X) as well.

Figure B.1 shows that an increase in the ownership-structure similarity has a larger impact when suppliers have an R&D investment above the average, consistent with the results of Freeman (2025) and Deng et al. (2025), who find that overlapping shareholders facilitate innovation only when investments hinge on the supplier side.

When the supplier is the trading partner mostly engaged in R&D, an increase in ownership structure similarity by one standard deviation raises the unconditional probability of an active trading partnership by 33.67% to 44.88%, depending on the specification. The effects seem slightly smaller when both companies invest heavily in R&D, with coefficients ranging from 28.27% to 41.41%. More importantly, notice that coefficients are positive when customers and suppliers have a below-average R&D intensity, ranging from 12.27% to 13.89%, suggesting that holdup cannot fully explain the role of overlapping shareholders. For further details of the estimations, see Table SA.4 in the Appendix.

Double marginalization. When customers and suppliers operate in less competitive markets, they apply successive markups to their marginal costs, leading to a problem known as *double marginalization*. Although not a general rule, double marginalization tends to reduce profits for all companies along the supply chain (Hamilton and Mqasqas, 1996), so firms often avoid it by employing downstream-profit revenue-sharing contracts and nonlinear pricing. In addition, this type of contract is attractive to companies

Figure B.1 – *The role of overlapping owners in alleviating holdup*



because it allows them to incorporate clauses that address the specificities of supply chain relationships, including product quality, technical specifications, retail services, and so on.

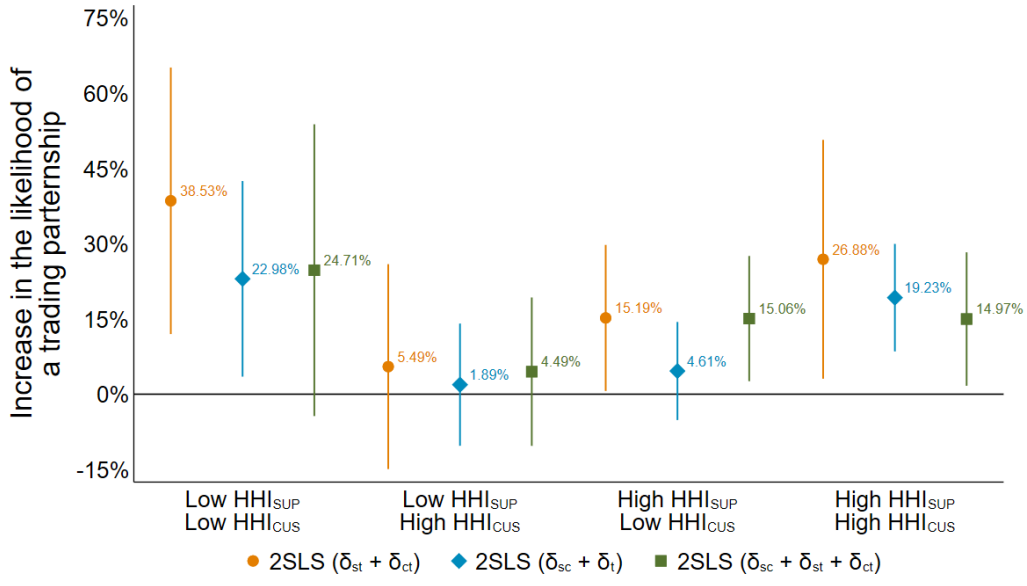
However, when not explicitly forbidden by competition authorities, these contracts are usually hard to enforce. In particular, because customers and suppliers need full information about each other's actions, which becomes increasingly demanding the more steps there are in the supply chain. Thus, as in the previous case, double marginalization can be a good example of contractual frictions in which overlapping shareholders can play a decisive role in overcoming it.

I proceed likewise as in the previous case, using market concentration as a rough indicator of competition. Although this often requires establishing a definition for product markets, I lean on a particular version of the Herfindahl-Hirschman Index (HHI) based on a text-based network industry classification (TNIC). Hoberg and Phillips (2016) constructs product differentiation distances across all publicly listed companiesin the US. Therefore, every other firm in the economy can be a potential competitor for a specific output produced by the firm.

The network approach makes it unnecessary to define yuxtaposed market definitions across industries, which is more appropriate in a setting with publicly listed companies in the US for two reasons. On the one hand, most of these companies are multi-product firms, so confining them to a single industry artificially inflates concentration measures within that market (though potentially creating biases in the opposite direction for other product markets). On the other hand, some companies may not have close substitutes within standard market classification—such as SIC or NAICS—, which can also introduce problems into the analysis. For more details on text-based product similarity and its properties, see Hoberg and Phillips (2016), Frésard et al. (2020), and Ederer and Pellegrino (2025).

I classify customers and suppliers based on whether their TNIC-HHI concentration is above or below the median (2,040 out of 10,000, which corresponds to moderate levels of concentration) among all firms in the sample. Namely, publicly listed US companies between 1999 and 2013, as described in Section I. I then extend Equation (X) to assess

Figure B.2 – *The role of overlapping owners in alleviating double marginalization*



heterogeneous effects of overlapping ownership across markets with different degrees of double marginalization.

$$Link_{sct} = \delta_{st} + \delta_{ct} + OvrOwn_{sct} \times \sum_{j \in \{L,H\}} \sum_{k \in \{L,H\}} \tau^{jk} \, (TNIC - HHI)^{jk} + \varepsilon_{sct} \quad \text{(B.2)}$$

where $(TNIC - HHI)^{LH}$ indicates that customer c faces relatively high competition, while supplier s holds a more dominant position. The same interpretation applies to the remaining dummy variables and the four coefficients τ^{LL}, τ^{LH}, τ^{HL}, and τ^{HH}. Likewise, I report results from extending Equation (X), following the same approach as above.

Figure B.2 illustrates several insightful results. First, overlapping shareholders play an extensive role when both the customer and the supplier hold a dominant position—which aligns with double marginalization—showing that an increase in the ownership-structure similarity increases the likelihood of an active trading partnership between 33.87% and 37.08%, depending on the specification. Second, when upstream and downstream markets exhibit greater competition, coefficients are larger, ranging from 47.43% to 47.88%. Contrary to the hypothesis of double marginalization, the last conclusion suggests that overlapping shareholders play a more significant role when customers and suppliers face more intense competition. For further details of the estimations, see Table SA.5 in the Appendix.

All in all, overlapping shareholders can help customers and suppliers overcome informational and contractual difficulties, overcoming problems that often require more cumbersome arrangements. Although the results support this conclusion for situations involving holdup and double marginalization, my findings do not rule out the possibility that overlapping owners play a significant role in other margins. That is, other contractual and informational frictions, as well as systemic risks affecting the supply chain. If anything, the effects of intense competition faced by customers and suppliers corroborate the importance of further exploring additional channels.

Nevertheless, these results add suggestive evidence to the literature, which has found

beneficial consequences of having overlapping owners on both sides of the supply chain relationship, with the potential to create value for trading partners, their shareholders, and the economy as a whole (Gao et al., 2022; Riva, 2022; Freeman, 2025; Deng et al., 2025). In particular, my findings propose that overlapping owners, beyond extending the duration of trading partnerships already in place, also facilitate the creation of supply chain relationships that might not have occurred otherwise.

C. Robustness (WIP)

The following section condenses all relevant robustness checks, each corresponding to a methodological decision described in the paper.

C.1. Alternative sample definitions

In the following subsection, I elaborate on the potential biases introduced by the wording of SFAS N° 131 and present robustness tests using four alternative criteria to define the sample. The first two sample definitions arise from discrepancies encountered in the two datasets that I use to define feasible and active trading partnerships. In contrast, the last two sample definitions rely on alternative approaches for determining the feasibility of each customer-supplier dyad.

Small customers. The wording of SFAS N° 131 can censor relatively small customers, even when these companies buy only from a few suppliers. Since these companies make smaller purchases, they become less likely to surpass the mandated 10% threshold. Furthermore, anticipating the direction of potential biases is not straightforward, as it depends on whether the ownership-structure similarity among these dyads lies above or below the current sample average.

Since 2SLS regressions estimate local effects—for this particular instrument and setting, that means customers and suppliers with more stable ownership structures over time—this source of bias does not affect the interpretation for the compliers in the sample.

If anything, the censorship could have introduced a downward bias in empirical exercises on double marginalization and holdup, since the missing trading partnerships would often be those in which a small customer requires a higher level of customization for its inputs or faces a monopolist supplier.

Discrepancies between feasible and active trading partnerships. When merging the datasets corresponding to feasible (Frésard et al., 2020) and active trading partnerships (Barrot and Sauvagnat, 2016), a few active supply chain relationships show a null vertical upstream relatedness. On the one hand, the first paper mentions that the procedure loses track of a small fraction of dyads when mapping CUSIP (American Bankers Association) and GVKEY (Compustat) identifiers. On the other hand, the second paper explains that using phonetic string-matching algorithms can lead to incongruent trading partnerships, even after manual review. Therefore, discrepancies arising when merging the datasets could indicate errors in one or both sources.

Throughout the paper, I lean on a compromise solution. First, I define all customer-supplier dyads in the dataset of active trading partnerships as feasible. Second, for supply chain relationships with a null vertical upstream relatedness, I assign them the average across active trading partnerships in the same year. For the remaining customer-supplier dyads, I retain the corresponding value reported in the dataset of feasible trading partnerships. To assess the importance of the assumption in my findings, I consider two distinct scenarios in which the error arises solely from one dataset.

Should inconsistencies arise from improper mapping between firms' names and GVKEY identifiers, I define the *Feasible-priority sample* by excluding all non-feasible customer-

supplier dyads, even if they appear in the dataset of active trading partnerships. Instead, if discrepancies arise in mapping vertical upstream relatedness between CUSIP and GVKEY identifiers, I define the *Active-priority sample* by assigning a weight equal to one to all active trading partnerships in the sample.

Alternative samples under different criteria. Alternatively, I identify a different set of feasible trading partnerships from patterns in the original supply chain network and the text-based network industry classification.

I denote the first definition as the *Network sample*, which requires me to blend all active trading partnerships between 1976 and 2013 into a unique supply chain network, a common practice in the literature on *link analysis*. After that, I proceed as follows. For any arbitrary customer c, I identify all its suppliers in the network and label the set as $\mathcal{S}_0$. Next, I retrieve all the customers of companies in $\mathcal{S}_0$ and tag them as $\mathcal{C}_1$. This step provides a set of closely related companies to c, even if they operate in different industries. Then, I retrieve all suppliers of companies in $\mathcal{C}_1$ and label them as $\mathcal{S}_1$. Finally, I define all companies in $\mathcal{S}_1$ as feasible suppliers of c. After repeating the procedure for every customer in the dataset of active trading partnerships, I retain annual observations in which companies simultaneously report book assets to the SEC between 1999 and 2013, in industries other than the financial and utility sectors.

The second definition, the *TNIC sample*, uses text-based product similarity to identify all competitors of customers and suppliers in the dataset of active trading partnerships (Hoberg and Phillips, 2016). As a result, every company j is assigned to a year-specific set of closely related companies, $\mathcal{P}_t^j$, in the product space. Then, for any given year t and customer c in the dataset, I consider all companies in $\mathcal{P}_t^c$ as potential customers of every observed supplier of c. Likewise, for any given supplier s, I define all companies in $\mathcal{P}_t^s$ as potential suppliers of every observed customer of s.

Robustness. Figure C.1 shows that conclusions remain unchanged across all alternative sample definitions. Quantitatively, a few patterns emerge that warrant mention.

First, differences across samples are sharper when using the specification with time-invariant customer and supplier fixed effects. This comes as no surprise, since this specification is more sensitive to supply chain persistence, which can significantly vary with the sample definition. This contributes to another reason for preferring interpretations based on time-varying customer and supplier fixed effects.

Second, coefficients for the *TNIC sample* are statistically significantly smaller. However, when compared to other definitions, the unconditional probability of an active trading partnership in this sample is 10 times higher, making the direct comparison less obvious. In any case, the sign and magnitude of the coefficients still point to similar conclusions as the remainder of the paper.

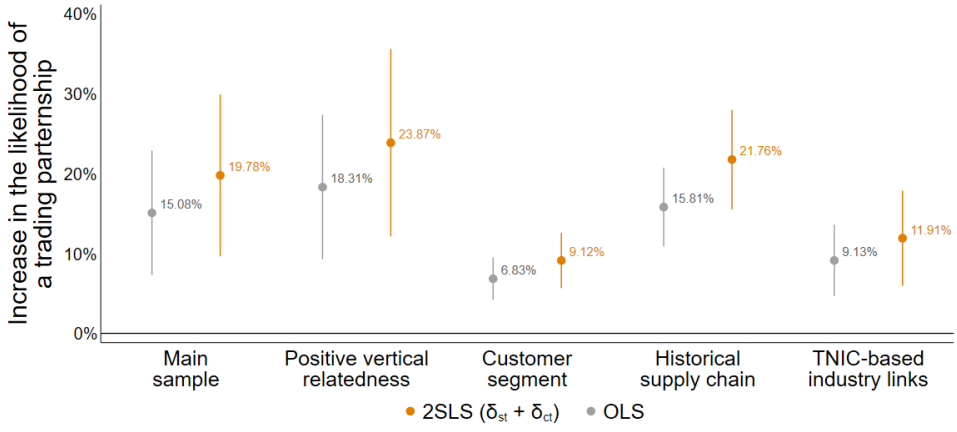
Third, the largest differences between OLS and 2SLS estimates occur in the *Active-priority sample*, where active trading partnerships carry much greater weight than the remaining customer-supplier dyads. This feature highlights an important aspect of the workarounds discussed in Section II.

On the one hand, the reweighting breaks the properties discussed in the section, as the approach no longer imposes *ex-ante* feasibility probabilities, leading to larger biases in the OLS estimates. On the other hand, the pattern shows that the IV strategy offers a solution even when feasibility is unobserved and unaccounted for, as it addresses biases arising from omitted variables.

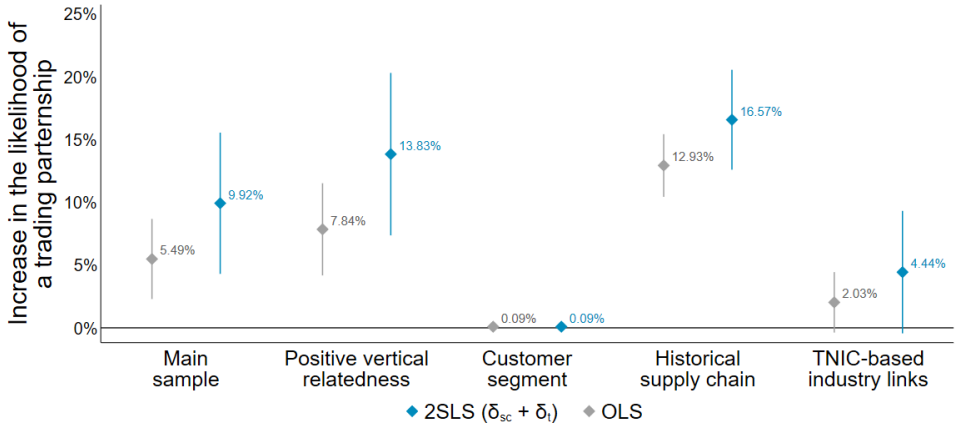
Taken together, and regardless of any disparities in the magnitude of the effects, all the specifications point to the same conclusion about the causal relationship between the ownership-structure similarity between customers and suppliers and the likelihood of an active trading partnership.

Figure C.1 – *Robustness: Alternative sample definitions*

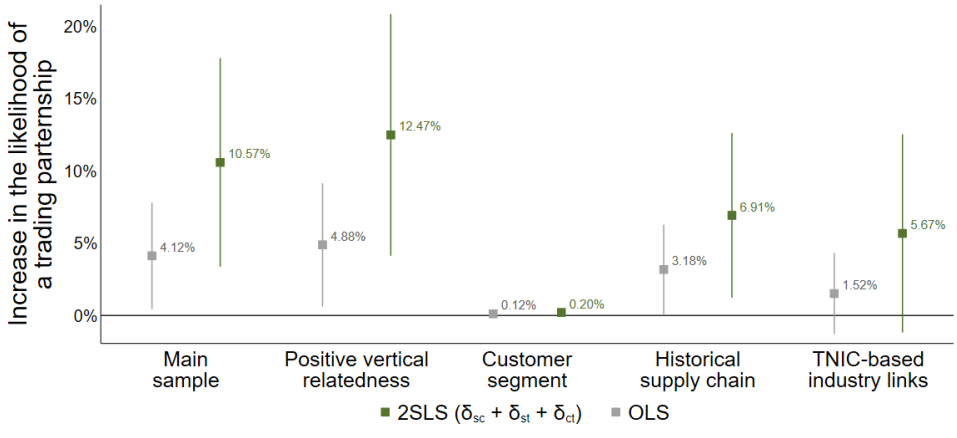
(a) FE-trend



(b) FE-dyad



(c) FE-triple



C.2. Alternative overlapping ownership measures

While my preferred measure of overlapping ownership between customers and suppliers relies on deriving profit weights, the literature offers a few other candidate metrics for measuring ownership structures and their similarities.

The following subsection tests the consistency of my findings using two standard measures found in prior studies. In addition, I propose a third variable that combines both profit weights into a single metric, which measures how likely it is for managers to agree on the benefits of engaging in a trading partnership.

Overlapping market value. Antón and Polk (2014) and Freeman (2025) are proponents of the following metric, which represents the fraction of the total market value of the supplier, V_{st}, and the customer, V_{ct}, that is owned by the set of overlapping shareholders $i \in \mathcal{O}$ in year t. Namely,

$$ovrMktVal_{sct} = \frac{\sum_{i \in \mathcal{O}} V_{sit} + V_{cit}}{V_{st} + V_{ct}}$$

Overlapping-shares product. Hansen and Lott (1996) and Freeman (2025) present the following measure, which is the cross-product of the customer's and supplier's shareholdings. That is,

$$ovrShrProd_{sct} = \sum_{i \in \mathcal{S}} \beta_{sit} \times \sum_{i \in \mathcal{S}} \beta_{cit}$$

Manager's agreeability. Last, I propose a metric that uses the smaller of the two profit weights, rather than the higher one used in Subsection III.B.

$$minKappa_{sct} = min(\kappa_{sct}, \kappa_{cst})$$

Intuitively, this metric should capture when both managers internalize the aggregate benefits of engaging in a trading partnership.

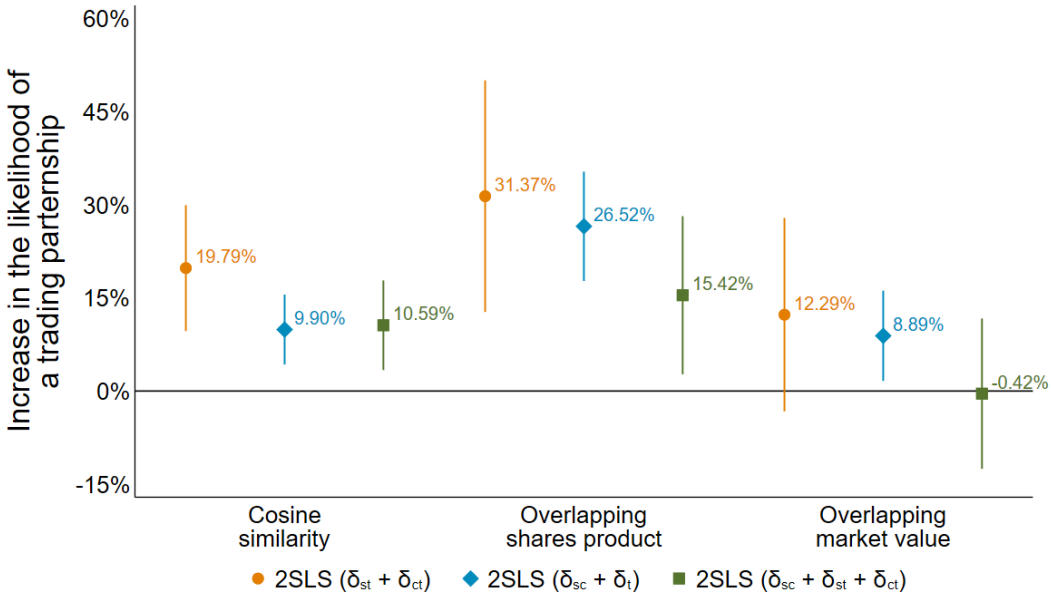
Robustness. Figure C.2 shows that coefficients are positive and statistically significant for all alternative metrics except for the overlapping market value, which is consistent with the findings in Freeman (2025). The author consistently reports coefficients for this metric that halve the effects identified by other alternatives. This is not a surprise, since the measure relies on companies' market capitalization in addition to their ownership structure, which introduces a further source of endogeneity that my instrumental variables approach does not address. As such, these results suffer from measurement error and identification problems, both of which bias the coefficients towards zero. For further details on the estimations, see Table SA.7 in the Appendix.

C.3. Alternative specifications for instruments

A potential threat to identification arises from the inherent nonlinearities in the ownership measures, which may capture unintended interactions between shifts in index weights and shareholdings. Throughout the paper, I present results using a linear approximation of the original expression (first-order Taylor expansion). In this subsection, I replicate the baseline analysis using the nonlinear functional form and a quadratic approximation (second-order Taylor expansion).

Robustness. Figure C.3 reveals that using the nonlinear functional form or any of the two approximations conveys almost identical quantitative results. In particular, coef-

Figure C.2 – *Robustness: Alternative overlapping ownership measures*



ficients obtained via approximations of the instrument are slightly smaller than those obtained using the nonlinear version. In conclusion, methodological choices regarding the instrument's functional form do not drive the patterns observed across the empirical exercises. For more details about the estimations, see SA.8 in the Appendix.

C.4. Robustness S&P 500 index

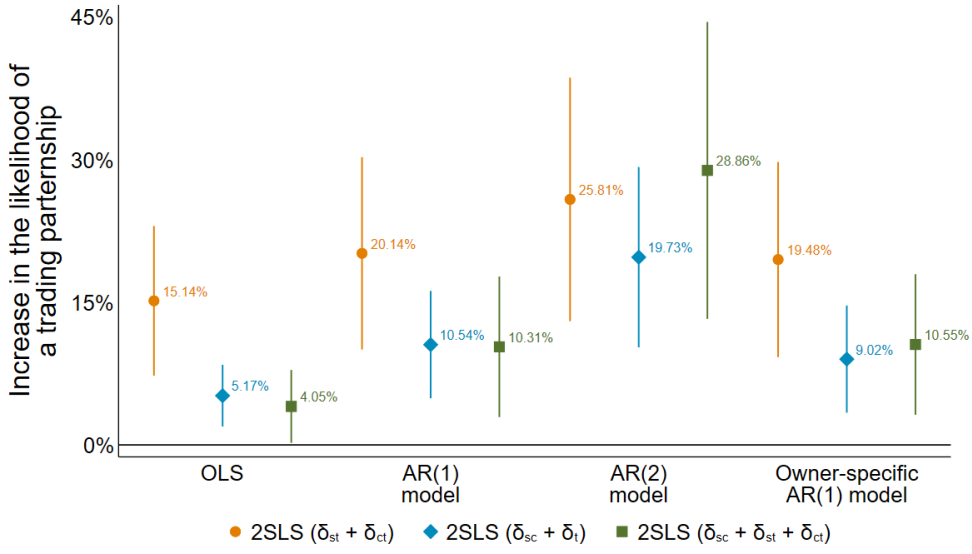
The premise is that index funds offer value to investors by tracking a set of sufficiently diversified assets, such as the S&P 500 index. Thus, upon changes in the index constituents, passive funds and other investors replicating the strategy must acquire equity in companies entering the index and sell the stocks of companies leaving it. In this section, I discuss the limitations of a few instruments in the literature and present a detailed description of the empirical approach used in the paper.

Additions and deletions in the literature. According to Lewellen and Lowry (2021), additions and deletions are inappropriate instruments for addressing the endogeneity of ownership structures. First, which companies make the cut into the index depends on a committee, whose decisions might reflect the companies' most recent performance.[25] For instance, committee members could be anticipating the trading partnerships of entering companies to improve, or the opposite for companies leaving the index. Second, being added to the index attracts the attention of the media, analysts, and a wider range of investors. As a result, the company's ownership structure can change through channels that are hard to isolate. For example, the authors note that companies recently added to indexes increase their institutional ownership while crowding out blockholders.

Intuitively, S&P 500 additions and deletions mechanically affect the weights of all remaining index constituents, given differences in the market values of entrants and

[25] According to the S&P 500 US Indices Methodology documentation: *"Constituent selection is at the discretion of the Index Committee and is based on the eligibility criteria. [...] Sector balance, [...] in the relevant market capitalization range, is also considered in the selection of companies for the indices."* It later adds, *"S&P Dow Jones Indices Index Committees reserve the right to make exceptions when applying the methodology if the need arises. In any scenario where the treatment differs from the general rules stated in this document or supplemental documents, clients will receive sufficient notice, whenever possible."*

Figure C.3 – *Robustness: Alternative instrument definitions*



exits. For instance, if the market value of entering companies were higher than that of exiting ones, index weights for the remaining companies would adjust proportionally downwards to make room for the higher index weights of the entrants. The opposite would hold as well.[26]

C.5. Constructing predicted shareholdings

Stable shareholders and companies. The empirical approach first requires dealing with additional potential sources of endogeneity. In particular, those that could influence the chain of effects linking additions and deletions in the S&P 500 to the companies' ownership structures. I drop 300,514 out of the 10,238,314 ownership ties from 2000 to 2013, involving companies in years when they participated—directly or indirectly—in an index addition or deletion. This includes not only entrant and exiting firms but also related parent companies, subsidiaries, spin-offs, and so on.

Furthermore, I leverage the fact that indexing[27] has become a widely used form of passive investing—with most institutional shareholders typically employing a diversified

[26] Because the index comprises the 500 most influential companies in the US according to S&P Indices, a typical change of its constituents would associate one addition with its corresponding deletion. However, publicly listed companies in the US engage in mergers, acquisitions, and spin-offs on a regular basis, prompting the S&P Indices Committee to respond when such transactions involve index constituents. Therefore, they do not always substitute low-market-capitalization companies for high-market-capitalization ones, making it challenging to anticipate the net change in the constituent's market value. For example, on December 14, 2011, S&P Indices announced that *TripAdvisor Inc. would replace Tellabs Inc. in the S&P 500 index by December 20, 2011. According to the press release, the announced day corresponds to the expected date on which the S&P 500 constituent Expedia was to complete the spin-off of TripAdvisor. Assuming a fixed number of outstanding shares and their prices, the market capitalization of Expedia before the spin-off should amount to the sum of TripAdvisor and Expedia during the first quarter of 2012. Therefore, this change in constituents only reflects the market value lost at Tellabs' exit. Conversely, on March 27, 2000, Standard & Poor's announced that Linear Technology Corp. and Pharmacia Corp. would replace Monsanto Company and Pharmacia & Upjohn in the S&P 500 index. The press release explains that Pharmacia is the merger of Monsanto and Pharmacia & Upjohn. Therefore, the change in constituents solely reflects the added market value from Linear Technology's entry. There are many other examples, with 51 spin-offs, 158 mergers and acquisitions, and several changes of names and tickers between 1999 and 2013. Therefore, γ should capture how institutional shareholders reallocate their passive investments within portfolios in response to shifts in index weights. For more details, Section* **??** *in the Appendix provides an extensive description of the use of additions and deletions from the S&P 500.*

[27] I refer to *indexing* as a strategy in which investors replicate the performance of specific market indexes, such as the S&P 500, by closely matching their holdings and weights to those of the index.

strategy that combines active and passive funds—and focus on shareholders with more stable positions within their portfolios. I exclude institutional investors purchasing shares in a company for the first time, which amounts to 3,248,053 additional ownership ties. For the purpose of computing ownership structures, I henceforth consider these investors as retail shareholders. That is, small individual investors who buy and sell company stock in tiny amounts.

S&P 500 index weighting.

Because the index lists the 500 most influential companies in the US according to S&P Indices, a typical change in constituents should involve replacing a low-market-capitalization company with a high-market-capitalization one. However, this is seldom the case. Publicly listed companies in the US regularly engage in mergers, acquisitions, and spin-offs, prompting the S&P Indices Committee to respond quickly and appropriately to such operations. As a result, the index does not always delete a company with a relatively low market value compared to the addition, making it challenging to anticipate the net change in the constituent companies' market values.

I manually collect news releases from Standard & Poor's, S&P Indices, and S&P Dow Jones Indices between 1999 and 2013. I select all announcements for additions and deletions related to the S&P 500, S&P MidCap 400, and S&P 600 SmallCap indexes. For the subset of S&P 500 constituents, I identify the exact date of each addition and deletion, the names and tickers of the companies involved, and a detailed description of the announcements. This way, I classify entering and exiting firms by whether the decision involves a change in the company's market capitalization, a merger, an acquisition, a spin-off, a renaming, or a delisting.

Shareholdings and index weights. To minimize endogeneity concerns arising from correlations between index weight determinants and other factors, such as outstanding shares and share prices, I set each firm's market capitalization to the quarter immediately preceding the corresponding addition or deletion. Furthermore, the S&P 500 belongs to the *float-adjusted market-capitalization-weighted indices* segment of S&P Dow Jones, which has two immediate implications. First, stocks with higher market capitalization have a greater impact on the index's performance than those with lower market capitalization. Second, market capitalization is based solely on floating outstanding shares. That is, it excludes shares held by long-term strategic shareholders, such as insiders, private equity firms, or the government, although I cannot make that distinction.[28]

As I mentioned before, it is important to exclude other companies related to the added and deleted firms. That group of companies is represented by $\mathcal{R}_t$, which includes parent companies, subsidiaries, spin-offs, and other companies explicitly mentioned in the announcement. In all cases, the explicit mention of a company in the announcements reveals that the Index Committee has considered its relationship with at least one of the additions and deletions when calling for a constituent change.

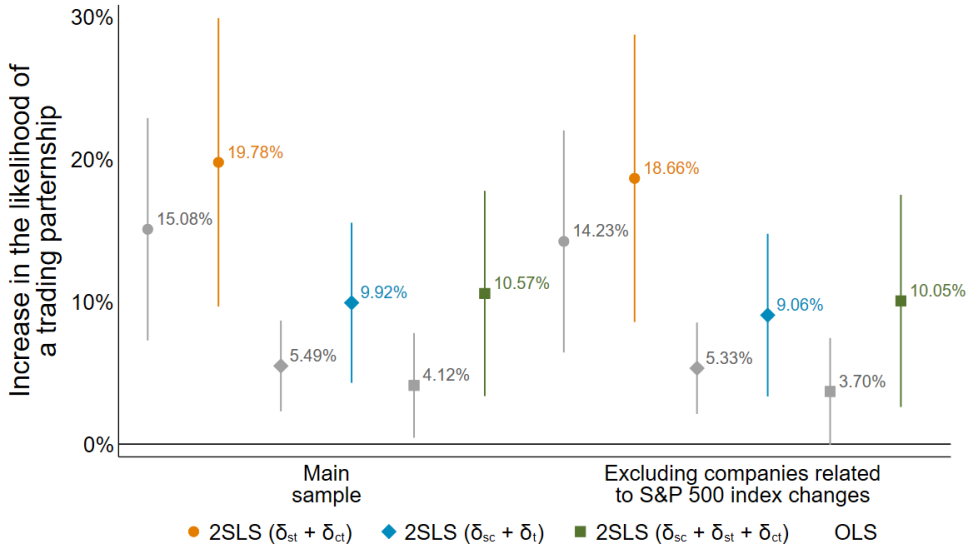
Predicted shareholdings. I estimate Equation (4) using OLS for the remaining 6,648,273 ownership ties in the sample. The estimation yields the following coefficients

$$(\alpha, \rho, \tau) = (0.0007, 0.9074, 0.0193)$$

all of which are statistically significant at the 99% confidence level using heteroskedastic-robust standard errors. More importantly, the R^2 is 0.815, suggesting that past shareholdings and index weight changes from additions and deletions to the S&P 500 capture

[28] For more information about the float adjustment methodology and investable weight factors, refer to `https://www.spglobal.com/spdji/en/documents/index-policies/methodology-sp-float-adjustment.pdf`.

Figure C.4 – *Robustness: Excluding additions, deletions, and companies related to changes in the constituents of the S&P 500 index*



most of the variation in portfolios.

The underlying assumption in the identification strategy is that, as long as the persistence in the supply chain declines faster than that of the ownership structure, the unexplained variation in this estimation captures most endogeneity and simultaneity in ownership ties. As such, I extend the same argument to the analogous versions of the ownership-structure similarity, the relative shareholder concentration, and the profit weights, which I construct using the predicted shareholdings from the estimated Equation (4).

To conclude, I am more inclined to use Equation (**??**) for estimating the 2SLS. As I mentioned above, the key determinant of the identification strategy is appropriately controlling for supply chain persistence. Thus, the inclusion of time-varying company fixed effects can do more to capture this persistence, as long as it is driven by certain customers and suppliers rather than idiosyncratic factors at the customer-supplier dyad level.

Robustness. Explanation

SUPPLEMENTARY MATERIAL

SA. Additional tables and figures

Table SA.1 – *Overlapping ownership increases the likelihood of active trading partnerships*

Dependent variable: Active trading partnership (Avg. likelihood: 0.0203%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Overlapping ownership	0.1508***	0.0549***	0.0412*				0.1978***	0.0992***	0.1057**
	(0.039834)	(0.016272)	(0.018758)				(0.051663)	(0.028684)	(0.036771)
Predicted overlapping ownership				0.6932***	0.5248***	0.4507***			
				(0.000166)	(0.000200)	(0.000214)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							17,482,474	6,879,157	4,435,221
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping ownership: 0.0926 (0.1369)

Table SA.2 – *More diffuse shareholdings amplify the effects of increasing the degree of overlapping ownership*

Dependent variable: Active trading partnership (Avg. likelihood: 0.0194%)

	OLS	FS			2SLS	OLS	FS			2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Overlapping ownership	0.3473***				0.4327***	0.1120***				0.1669***
	(0.066766)				(0.092181)	(0.028270)				(0.046134)
Deviation from identical concentration	0.1920**				0.2236*	0.0278				0.0136
	(0.074325)				(0.102901)	(0.032560)				(0.065830)
Profit weight	0.3712***				0.4352**	0.1017*				0.1209
	(0.087071)				(0.136447)	(0.042749)				(0.079074)
Predicted overlapping ownership		0.7232***	-0.0395***	-0.1687***			0.5801***	-0.0239***	-0.1473***	
		(0.000207)	(0.000101)	(0.000225)			(0.000238)	(0.000113)	(0.000196)	
Predicted deviation from identical concentration		0.0238***	0.2431***	0.0121***			0.0562***	0.1229***	0.0182***	
		(0.000141)	(0.000213)	(0.000188)			(0.000148)	(0.000110)	(0.000122)	
Predicted profit weight		0.0497***	-0.0008***	0.2145***			0.0841***	0.0079***	0.1414***	
		(0.000224)	(0.000128)	(0.000286)			(0.000252)	(0.000120)	(0.000219)	
N	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486
F-statistic		1,096,108	1,496,108	859,111	318,848		856,803	1,390,531	921,901	369,106
Company × Time FE	✓	✓	✓	✓	✓					
Dyad FE						✓	✓	✓	✓	✓
Time FE						✓	✓	✓	✓	✓

Overlapping ownership: 0.1087 (0.1428) Deviation from identical concentration: 0.7800 (0.9365)

Table SA.2 – *More diffuse shareholdings amplify the effects of increasing the degree of overlapping ownership* (continued...)

Dependent variable: Active trading partnership (Avg. likelihood: 0.0194%)

	OLS	FS			2SLS
	(11)	(12)	(13)	(14)	(15)
Overlapping ownership	0.0921**				0.1864***
	(0.029801)				(0.056602)
Deviation from identical concentration	-0.0296				-0.0194
	(0.039652)				(0.083329)
Profit weight	0.1056*				0.1397
	(0.043974)				(0.096101)
Predicted overlapping ownership		0.4972***	-0.0247***	-0.1236***	
		(0.000247)	(0.000101)	(0.000212)	
Predicted deviation from identical concentration		0.0272***	0.1576***	0.0037***	
		(0.000142)	(0.000180)	(0.000154)	
Predicted profit weight		0.0613***	-0.0083***	0.1323***	
		(0.000205)	(0.000109)	(0.000212)	
N	117,316,486	117,316,486	117,316,486	117,316,486	117,316,486
F-statistic		806,497	923,218	611,425	198,363
Company × Time FE	✓	✓	✓	✓	✓
Dyad FE	✓	✓	✓	✓	✓
Time FE					

Overlapping ownership: 0.1087 (0.1428) Deviation from identical concentration: 0.7800 (0.9365)

Table SA.3 – *Tunneling effects are larger but concentrated among dyads with diffuse-ownership suppliers*

Dependent variable: Active trading partnership (Avg. likelihood: 0.0205%)

	Sub-sample: Dyads with relatively more diluted suppliers					Sub-sample: Dyads with relatively more diluted customers				
	OLS	FS			2SLS	OLS	FS			2SLS
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Overlapping ownership	0.0725				0.0676	0.5839***				0.7547***
	(0.058880)				(0.091128)	(0.121426)				(0.162328)
Deviation from identical concentration										
Profit weight	-0.0671				-0.1706	0.7732***				0.9537***
	(0.090129)				(0.151046)	(0.157009)				(0.228498)
Predicted overlapping ownership		0.7029***		-0.1635***			0.7026***		-0.1609***	
		(0.000312)		(0.000343)			(0.000314)		(0.000344)	
Predicted deviation from identical concentration										
Predicted profit weight		0.0313***		0.2566***			0.0283***		0.2611***	
		(0.000370)		(0.000510)			(0.000370)		(0.000506)	
N	55,481,190	55,481,190		55,481,190	55,481,190	55,482,039	55,482,039		55,482,039	55,482,039
F-statistic		557,866		316,209	151,608		560,997		321,527	152,562
Company × Time FE	✓	✓		✓	✓	✓	✓		✓	✓
Dyad FE										
Time FE										

Overlapping ownership: 0.1180 (0.1468) Deviation from identical concentration: 0.7800 (0.9365)

Table SA.3 – *Tunneling effects are larger but concentrated among dyads with diffuse-ownership suppliers* (continued...)

Dependent variable: Active trading partnership (Avg. likelihood: 0.0205%)

	Sub-sample: Dyads with relatively more diluted suppliers					Sub-sample: Dyads with relatively more diluted customers				
	OLS	FS			2SLS	OLS	FS			2SLS
	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
Overlapping ownership	-0.0402				-0.0801	0.2631***				0.4969***
	(0.027171)				(0.056024)	(0.051117)				(0.101193)
Deviation from identical concentration	-0.0618				-0.1330	0.0967				0.3282*
	(0.033972)				(0.104915)	(0.055455)				(0.133000)
Profit weight	-0.0707				-0.1721	0.2173**				0.4791**
	(0.041649)				(0.107647)	(0.069645)				(0.155861)
Predicted overlapping ownership		0.5197***	-0.0182***	-0.1392***			0.5182***	-0.0162***	-0.1375***	
		(0.000356)	(0.000171)	(0.000279)			(0.000356)	(0.000171)	(0.000282)	
Predicted deviation from identical concentration		0.0257***	0.1158***	0.0232***			0.0229***	0.1160***	0.0261***	
		(0.000243)	(0.000179)	(0.000209)			(0.000240)	(0.000174)	(0.000208)	
Predicted profit weight		0.0318***	0.0115***	0.1400***			0.0278***	0.0134***	0.1434***	
		(0.000371)	(0.000194)	(0.000341)			(0.000368)	(0.000192)	(0.000338)	
N	55,481,190	55,481,190	55,481,190	55,481,190	55,481,190	55,482,039	55,482,039	55,482,039	55,482,039	55,482,039
F-statistic		269,780	504,964	310,175	133,450		269,887	514,972	317,982	134,049
Company × Time FE										
Dyad FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Overlapping ownership: 0.1180 (0.1468) Deviation from identical concentration: 0.7800 (0.9365)

Table SA.3 – *Tunneling effects are larger but concentrated among dyads with diffuse-ownership suppliers* (continued...)

Dependent variable: Active trading partnership (Avg. likelihood: 0.0205%)

	Sub-sample: Dyads with relatively more diluted suppliers					Sub-sample: Dyads with relatively more diluted customers				
	OLS	FS			2SLS	OLS	FS			2SLS
	(21)	(22)	(23)	(24)	(25)	(26)	(27)	(28)	(29)	(30)
Overlapping ownership	-0.0427				-0.0796	0.2060***				0.4341***
	(0.033343)				(0.084216)	(0.053756)				(0.114567)
Deviation from identical concentration										
Profit weight	-0.0813				-0.2865*	0.2431**				0.4940**
	(0.043930)				(0.142040)	(0.074910)				(0.187540)
Predicted overlapping ownership		0.4431***		-0.1143***			0.4425***		-0.1126***	
		(0.000374)		(0.000319)			(0.000376)		(0.000320)	
Predicted deviation from identical concentration										
Predicted profit weight		0.0207***		0.1391***			0.0184***		0.1419***	
		(0.000338)		(0.000375)			(0.000338)		(0.000374)	
N	55,481,190	55,481,190		55,481,190	55,481,190	55,482,039	55,482,039		55,482,039	55,482,039
F-statistic		223,688		178,037	82,705		225,390		179,596	82,911
Company × Time FE	✓	✓		✓	✓	✓	✓		✓	✓
Dyad FE	✓	✓		✓	✓	✓	✓		✓	✓
Time FE										

Overlapping ownership: 0.1180 (0.1468) Deviation from identical concentration: 0.7800 (0.9365)

Table SA.4 – *The role of overlapping owners in alleviating holdup*

Dependent variable: Active trading partnership (Avg. likelihood, pooled: 0.0200%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — $R\&D_{SUP}^{Low}$	$\&R\&D_{CUS}^{Low}$								
Overlapping ownership	0.1046	0.0566*	0.0393				0.1433	0.1009*	0.1068*
	(0.061990)	(0.023920)	(0.027538)				(0.078830)	(0.040401)	(0.051872)
Predicted overlapping ownership				0.7080***	0.5470***	0.4708***			
				(0.000225)	(0.000269)	(0.000293)			
N	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798	73,904,798
F-statistic							9,928,030	4,128,420	2,573,284
Panel B — $R\&D_{SUP}^{Low}$	$\&R\&D_{CUS}^{High}$								
Overlapping ownership	-0.0148	0.0266	0.0344				-0.0275	0.0519	0.0688
	(0.029359)	(0.017487)	(0.021335)				(0.040149)	(0.030954)	(0.042832)
Predicted overlapping ownership				0.6719***	0.5017***	0.4298***			
				(0.000366)	(0.000440)	(0.000473)			
N	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430	35,648,430
F-statistic							3,366,960	1,302,956	825,772
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping ownership (pooled): 0.0931 (0.1372)

Table SA.4 – *The role of overlapping owners in alleviating holdup* (continued...)

Dependent variable: Active trading partnership (Avg. likelihood, pooled: 0.0200%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — $R\&D_{SUP}^{High}$ &$R\&D_{CUS}^{Low}$									
Overlapping ownership	0.4092***	0.0869*	0.0636				0.5191***	0.1489	0.1566
	(0.099125)	(0.043832)	(0.054249)				(0.132276)	(0.080443)	(0.109241)
Predicted overlapping ownership				0.6731***	0.5032***	0.4309***			
				(0.000363)	(0.000428)	(0.000463)			
N	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661	35,649,661
F-statistic							3,430,675	1,378,944	865,139
Panel D — $R\&D_{SUP}^{High}$ &$R\&D_{CUS}^{High}$									
Overlapping ownership	0.3650***	0.0163	-0.0055				0.5143***	0.0353	0.0219
	(0.096574)	(0.049516)	(0.056679)				(0.132706)	(0.095072)	(0.122865)
Predicted overlapping ownership				0.6485***	0.4613***	0.3999***			
				(0.000576)	(0.000672)	(0.000729)			
N	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427	17,180,427
F-statistic							1,266,765	471,746	301,301
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping ownership (pooled): 0.0931 (0.1372)

Table SA.5 – *The role of overlapping owners in alleviating double marginalization*

Dependent variable: Active trading partnership (Avg. likelihood, pooled: 0.0189%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — HHI_{SUP}^{Low} &HHI_{CUS}^{Low}									
Overlapping ownership	0.2874**	0.1181*	0.0888				0.3853**	0.2298*	0.2471
	(0.104545)	(0.055526)	(0.074258)				(0.135495)	(0.099433)	(0.148332)
Predicted overlapping ownership				0.6844***	0.5216***	0.4448***			
				(0.000373)	(0.000447)	(0.000473)			
N	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564	26,428,564
F-statistic							3,370,000	1,359,910	885,384
Panel B — HHI_{SUP}^{Low} &HHI_{CUS}^{High}									
Overlapping ownership	0.0495	0.0068	0.0039				0.0549	0.0189	0.0449
	(0.078761)	(0.036827)	(0.042301)				(0.104168)	(0.062114)	(0.075443)
Predicted overlapping ownership				0.6972***	0.5408***	0.4655***			
				(0.000339)	(0.000387)	(0.000423)			
N	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266	35,524,266
F-statistic							4,236,746	1,952,579	1,208,645
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping ownership (pooled): 0.0990 (0.140235)

Table SA.5 – *The role of overlapping owners in alleviating double marginalization* (continued...)

Dependent variable: Active trading partnership (Avg. likelihood, pooled: 0.0189%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — HHI_{SUP}^{High} &HHI_{CUS}^{Low}									
Overlapping ownership	0.0797	0.0239	0.0582				0.1519*	0.0461	0.1506*
	(0.060843)	(0.027433)	(0.032561)				(0.074252)	(0.049858)	(0.063660)
Predicted overlapping ownership				0.6966***	0.5395***	0.4655***			
				(0.000335)	(0.000383)	(0.000418)			
N	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229	35,524,229
F-statistic							4,323,690	1,982,176	1,239,393
Panel D — HHI_{SUP}^{High} &HHI_{CUS}^{High}									
Overlapping ownership	0.2140*	0.1221***	0.0669				0.2688*	0.1923***	0.1497*
	(0.097305)	(0.033570)	(0.036831)				(0.121421)	(0.054654)	(0.067832)
Predicted overlapping ownership				0.7095***	0.5518***	0.4829***			
				(0.000290)	(0.000328)	(0.000366)			
N	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511	47,740,511
F-statistic							5,980,651	2,827,327	1,736,939
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping ownership (pooled): 0.0990 (0.140235)

Table SA.6 – *Robustness: Alternative sample definitions*

Dependent variable: Active trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — Main sample									
Overlapping ownership	0.1508***	0.0549***	0.0412*				0.1978***	0.0992***	0.1057**
	(0.039834)	(0.016272)	(0.018758)				(0.051663)	(0.028684)	(0.036771)
Predicted overlapping ownership				0.6932***	0.5248***	0.4507***			
				(0.000166)	(0.000200)	(0.000214)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							17,482,474	6,879,157	4,435,221
Avg. Overlapping ownership: 0.0926 (0.1369) Avg. likelihood: 0.0203%									
Panel B — Positive vertical relatedness									
Overlapping ownership	0.1831***	0.0784***	0.0488*				0.2387***	0.1383***	0.1247**
	(0.046167)	(0.018719)	(0.021723)				(0.059874)	(0.033000)	(0.042608)
Predicted overlapping ownership				0.6932***	0.5248***	0.4507***			
				(0.000167)	(0.000203)	(0.000216)			
N	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516	164,847,516
F-statistic							17,230,559	6,697,452	4,369,500
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. Overlapping ownership: 0.0926 (0.1369) Avg. likelihood: 0.0175%									

Table SA.6 – *Robustness: Alternative sample definitions* (continued...)

Dependent variable: Active trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — Customer segment									
Overlapping ownership	0.0683***	0.0009	0.0012*				0.0912***	0.0009	0.0020
	(0.013591)	(0.000476)	(0.000556)				(0.017613)	(0.000832)	(0.001083)
Predicted overlapping ownership				0.6959***	0.5237***	0.4501***			
				(0.000306)	(0.000326)	(0.000292)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							5,163,283	2,573,414	2,374,504
	Avg. Overlapping ownership: 0.0938 (0.1384) Avg. likelihood: 2.3678%								
Panel D — Historical supply chain									
Overlapping ownership	0.1581***	0.1293***	0.0318*				0.2176***	0.1657***	0.0691*
	(0.025226)	(0.012728)	(0.015766)				(0.031771)	(0.020303)	(0.029003)
Predicted overlapping ownership				0.7441***	0.6039***	0.5217***			
				(0.000554)	(0.000663)	(0.000731)			
N	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597	5,930,597
F-statistic							1,802,470	828,394	509,762
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
	Avg. Overlapping ownership: 0.0943 (0.1507) Avg. likelihood: 0.4555%								

Table SA.6 – *Robustness: Alternative sample definitions* (continued...)

Dependent variable: Active trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel E — TNIC-based industry links									
Overlapping ownership	0.0913***	0.0203	0.0152				0.1191***	0.0444	0.0567
	(0.022847)	(0.012270)	(0.014308)				(0.030429)	(0.024877)	(0.034927)
Predicted overlapping ownership				0.7047***	0.4685***	0.3872***			
				(0.001076)	(0.001591)	(0.001698)			
N	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444	1,432,444
F-statistic							429,253	86,735	52,005
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping ownership: 0.1147 (0.1608) Avg. likelihood: 1.7399%

Table SA.7 – *Robustness: Alternative overlapping ownership measures*

Dependent variable: Active trading partnership (Avg. likelihood: 0.0203%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — Cosine similarity									
Overlapping ownership	0.1509***	0.0549***	0.0412*				0.1979***	0.0990***	0.1059**
	(0.039860)	(0.016281)	(0.018773)				(0.051699)	(0.028701)	(0.036810)
Predicted overlapping ownership				0.6933***	0.5248***	0.4507***			
				(0.000167)	(0.000202)	(0.000215)			
N	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470
F-statistic							17,297,004	6,735,506	4,378,352
Avg. Cosine similarity: 0.0926 (0.136929)									
Panel B — Overlapping-shares product									
Overlapping ownership	0.2627***	0.1108***	0.0585				0.3137***	0.2652***	0.1542*
	(0.054158)	(0.027412)	(0.034024)				(0.095138)	(0.044908)	(0.064937)
Predicted overlapping ownership				0.9778***	0.8567***	0.7694***			
				(0.000771)	(0.000343)	(0.000390)			
N	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470
F-statistic							1,606,784	6,252,289	3,900,807
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. Overlapping-shares product: 0.0547 (0.092987)									

Table SA.7 – *Robustness: Alternative overlapping ownership measures* (continued...)

Dependent variable: Active trading partnership (Avg. likelihood: 0.0203%)

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel C — Overlapping market value									
Overlapping ownership	0.0366	-0.0159	-0.0610				0.1229	0.0889*	-0.0042
	(0.051204)	(0.023450)	(0.033950)				(0.079439)	(0.037076)	(0.061734)
Predicted overlapping ownership				0.8937***	0.7268***	0.6322***			
				(0.000178)	(0.000199)	(0.000205)			
N	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470	164,772,470
F-statistic							25,155,483	13,365,227	9,530,321
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	

Avg. Overlapping market value: 0.1610 (0.172495)

Table SA.8 – *Robustness: Alternative instrument definitions*

Dependent variable: Active trading partnership (Avg. likelihood: 0.0214%)

	OLS	AR(1) model		AR(2) model		Owner-specific AR(1) model	
	(1) OLS	(2) FS	(3) 2SLS	(4) FS	(5) 2SLS	(6) FS	(7) 2SLS
Overlapping ownership	0.1514***		0.2014***		0.2581***		0.1948***
	(0.040111)		(0.051576)		(0.065317)		(0.052329)
Predicted overlapping ownership		0.6966***		0.4945***		0.6711***	
		(0.000166)		(0.000212)		(0.000170)	
F-statistic			17,518,434		5,418,522		15,565,732
Company × Time FE	✓	✓	✓	✓	✓	✓	✓
Overlapping ownership	0.0517**		0.1054***		0.1973***		0.0902**
	(0.016531)		(0.028795)		(0.048355)		(0.028717)
Predicted overlapping ownership		0.5265***		0.2769***		0.4874***	
		(0.000209)		(0.000219)		(0.000200)	
F-statistic							
Dyad FE	✓	✓	✓	✓	✓	✓	✓
Time FE	✓	✓	✓	✓	✓	✓	✓
Overlapping ownership	0.0405*		0.1031**		0.2886***		0.1055**
	(0.019549)		(0.037706)		(0.079628)		(0.037686)
Predicted overlapping ownership		0.4521***		0.1838***		0.4443***	
		(0.000223)		(0.000218)		(0.000224)	
F-statistic							
Company × Time FE	✓	✓	✓	✓	✓	✓	✓
Dyad FE	✓	✓	✓	✓	✓	✓	✓
N	142,691,158	142,691,158	142,691,158	142,691,158	142,691,158	142,691,158	142,691,158

Avg. Overlapping ownership: 0.0990 (0.1405)

Table SA.9 – *Robustness: Excluding additions, deletions, and companies related to changes in the constituents of the S&P 500 index*

Dependent variable: Active trading partnership

	OLS			FS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A — Main sample									
Overlapping ownership	0.1508***	0.0549***	0.0412*				0.1978***	0.0992***	0.1057**
	(0.039834)	(0.016272)	(0.018758)				(0.051663)	(0.028684)	(0.036771)
Predicted overlapping ownership				0.6932***	0.5248***	0.4507***			
				(0.000166)	(0.000200)	(0.000214)			
N	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088	164,853,088
F-statistic							17,482,474	6,879,157	4,435,221
Avg. Overlapping ownership: 0.0926 (0.1369) Avg. likelihood: 0.0203%									
Panel B — Excluding companies related to S&P 500 index changes									
Overlapping ownership	0.1423***	0.0533**	0.0370				0.1866***	0.0906**	0.1005**
	(0.039763)	(0.016368)	(0.019147)				(0.051479)	(0.029140)	(0.038004)
Predicted overlapping ownership				0.6915***	0.5252***	0.4502***			
				(0.000172)	(0.000207)	(0.000222)			
N	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494	158,229,494
F-statistic							16,230,683	6,451,777	4,105,801
Company × Time FE	✓		✓	✓		✓	✓		✓
Dyad FE		✓	✓		✓	✓		✓	✓
Time FE		✓			✓			✓	
Avg. Overlapping ownership: 0.0907 (0.1351) Avg. likelihood: 0.0187%									